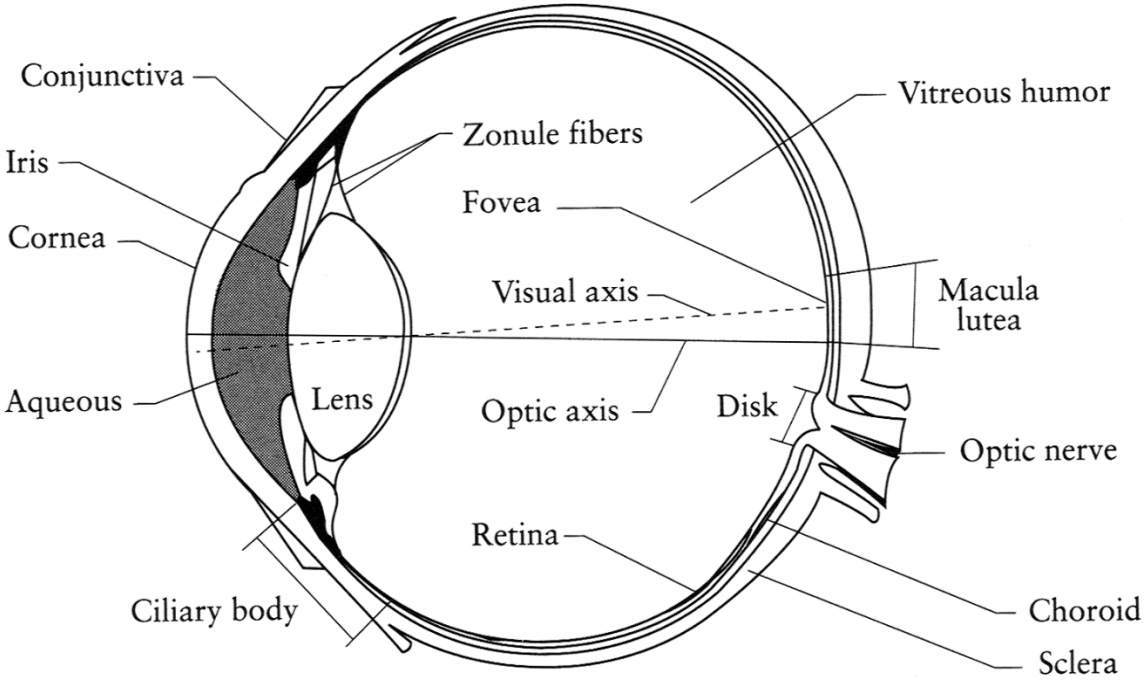
Computer Vision
Lecture 2

桑农 王岳环

Slides from CV course with Prof. Trevor Darrel, James Hayes

The Eye



The human eye is a camera!

Iris - colored annulus with radial muscles

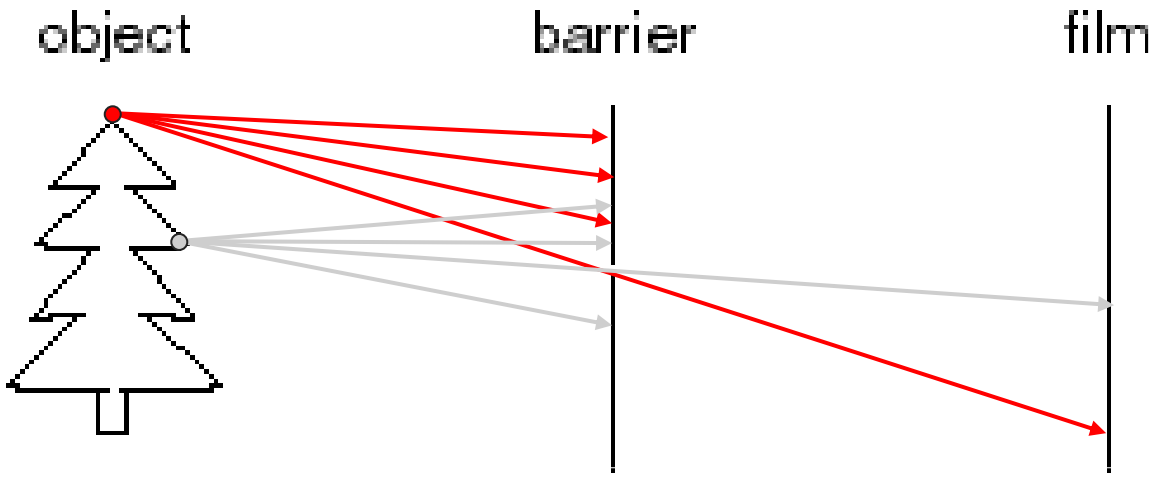
Pupil - the hole (aperture) whose size is controlled by the iris

What's the "film"?

– photoreceptor cells (rods and cones) in the **retina**

Slide by Steve Seitz

Pinhole camera

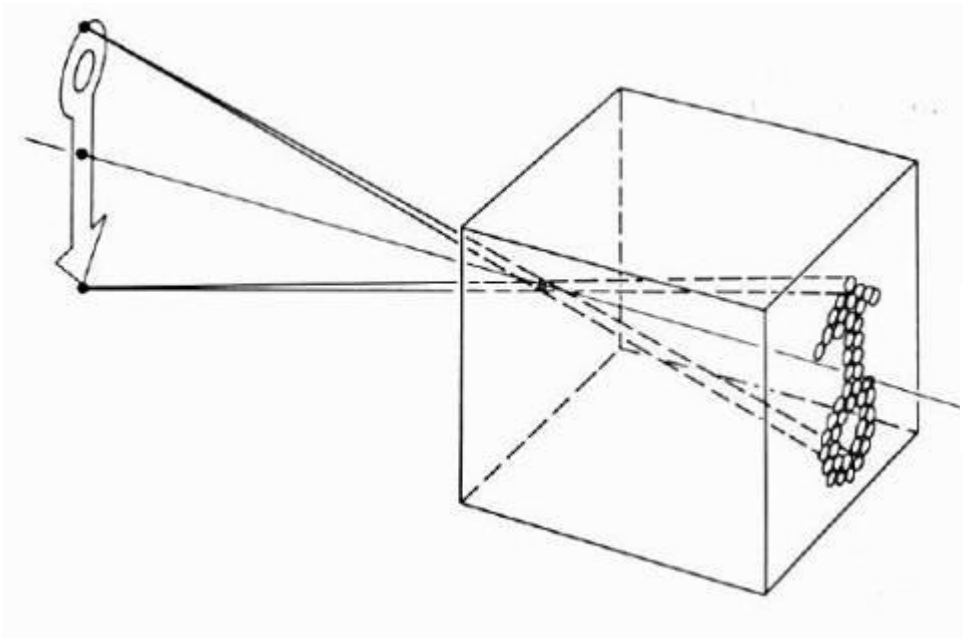


Add a barrier to block off most of the rays

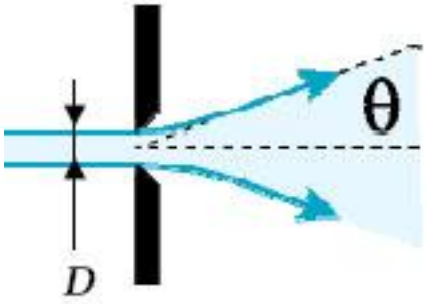
- This reduces blurring
- The opening is known as the **aperture**

Slide by Steve Seitz

Pinhole camera model



$$\theta = 1.22\frac{\lambda}{D}$$



Pinhole model:

Captures **pencil of rays** – all rays through a single point

The point is called **Center of Projection (focal point)**

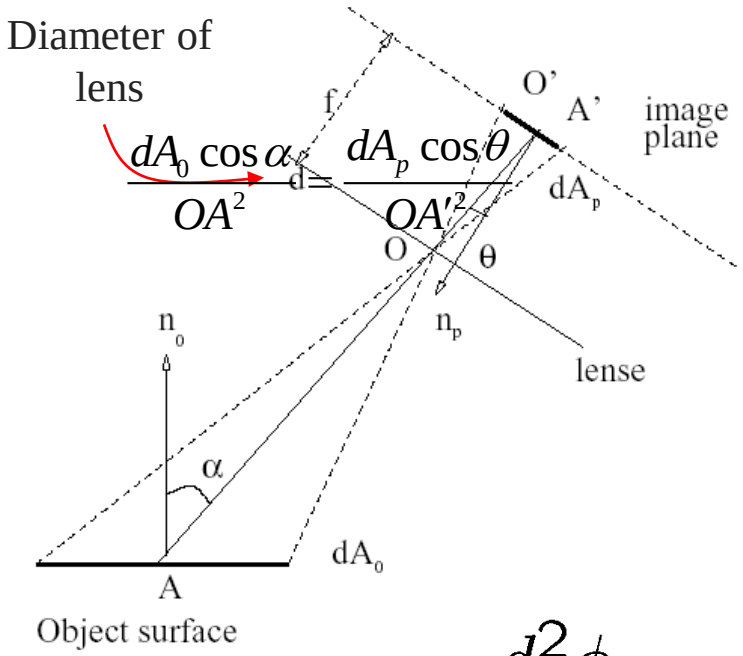
The image is formed on the **Image Plane**

Slide by Steve Seitz

Shrinking the aperture



The Relationship Between Image Intensity and Object Radiance



The solid angle for the entire lens is

$$dw_0 = \frac{\left(\pi d^2/4\right)\cos\theta}{r^2}$$

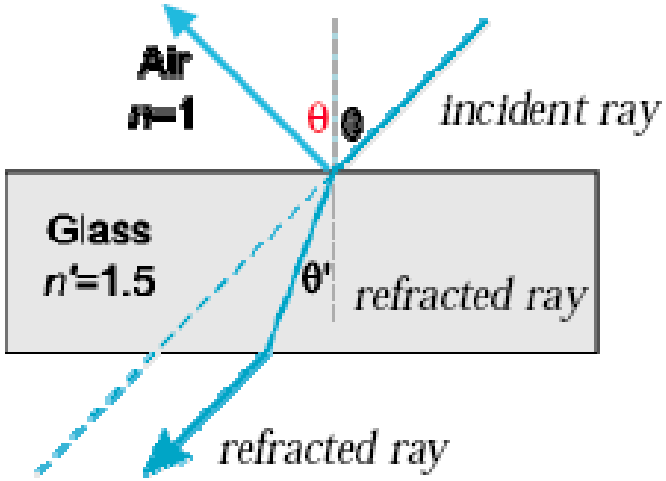
The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha dw_0$$

$$= L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

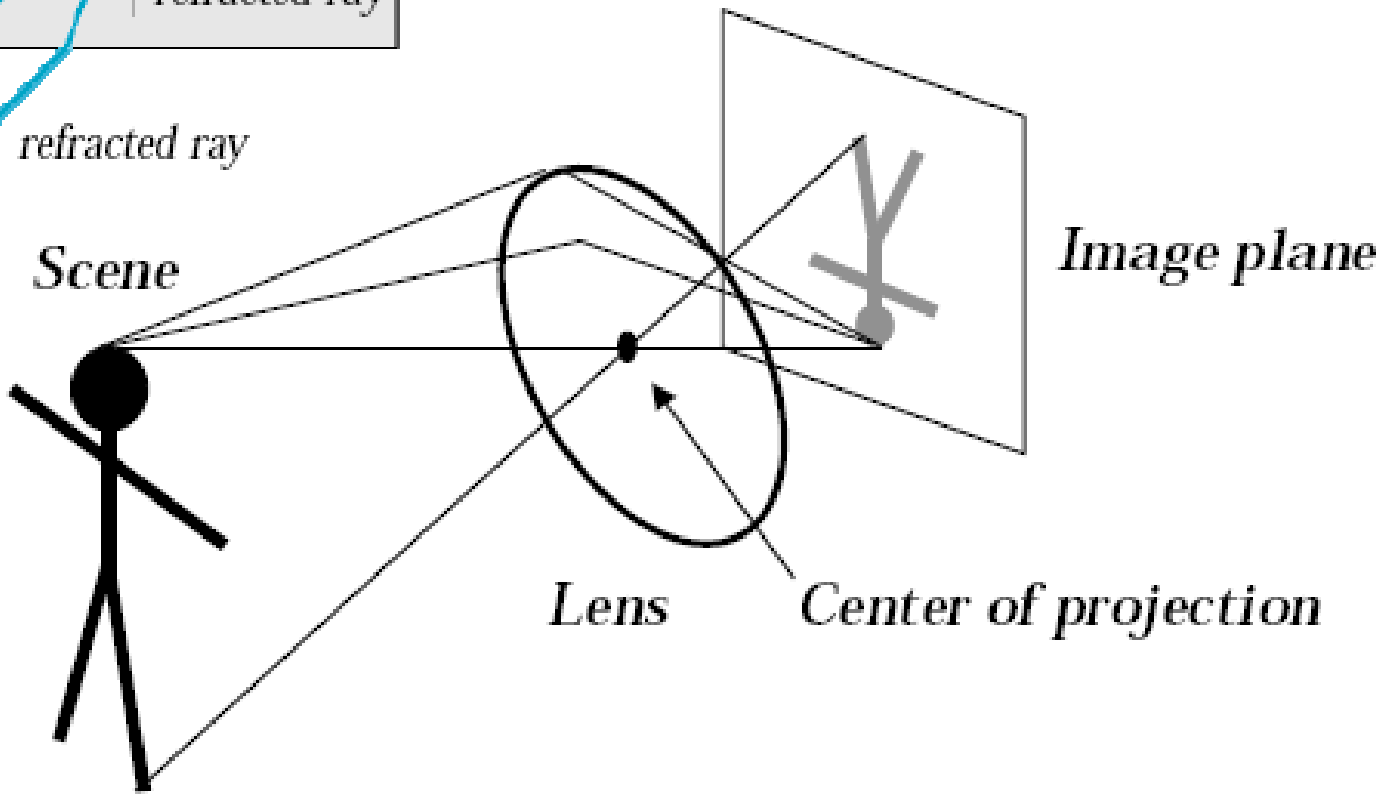
$$E = \frac{d^2\phi_r}{dA_p} = \frac{\pi}{4}\left(\frac{D}{f}\right)^2 cos^4\theta L_r$$

Function of Lenses



Snell's Law:

$$n\sin\theta = n'\sin\theta'$$



Thin lens formula——Paraxial optics

$$\frac{s^2}{n^2(s+r)^2} - \frac{s'^2}{n'^2(s'-r)^2} = -4r\sin^2(\frac{\phi}{2})$$

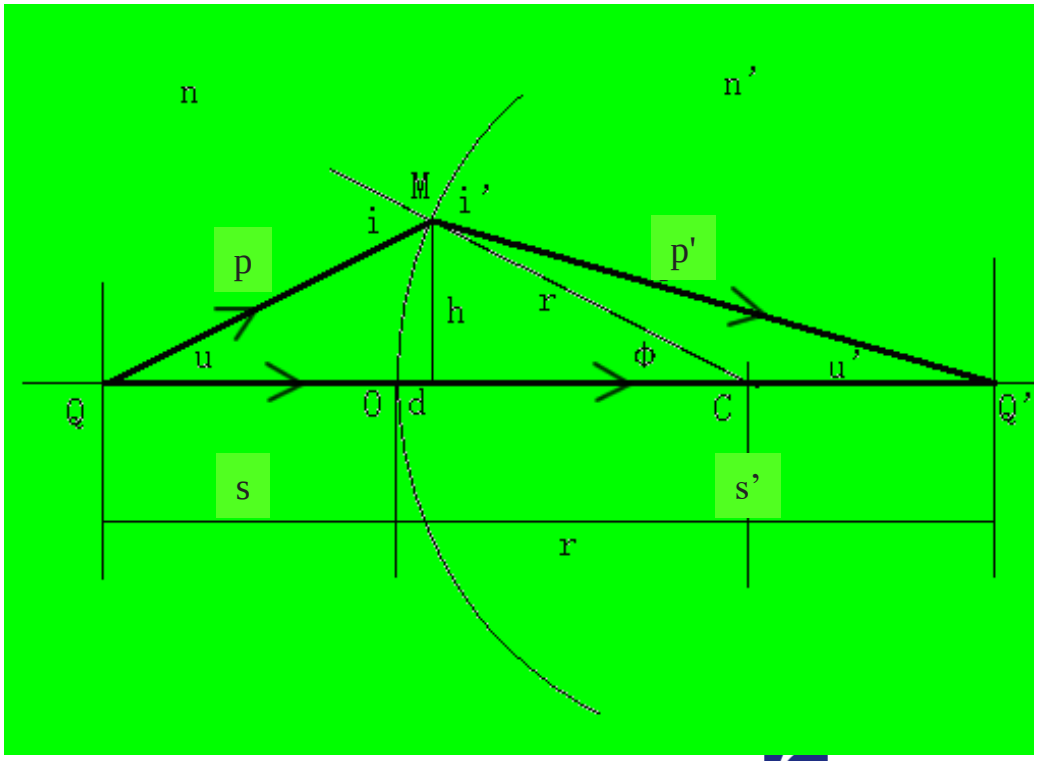
diopter

$$\rightarrow \quad \frac{n(s+r)}{s} = \frac{n'(s'-r)}{s'} \quad \rightarrow \quad \frac{n'}{s'} + \frac{n}{s} = \frac{n'-n}{r}$$

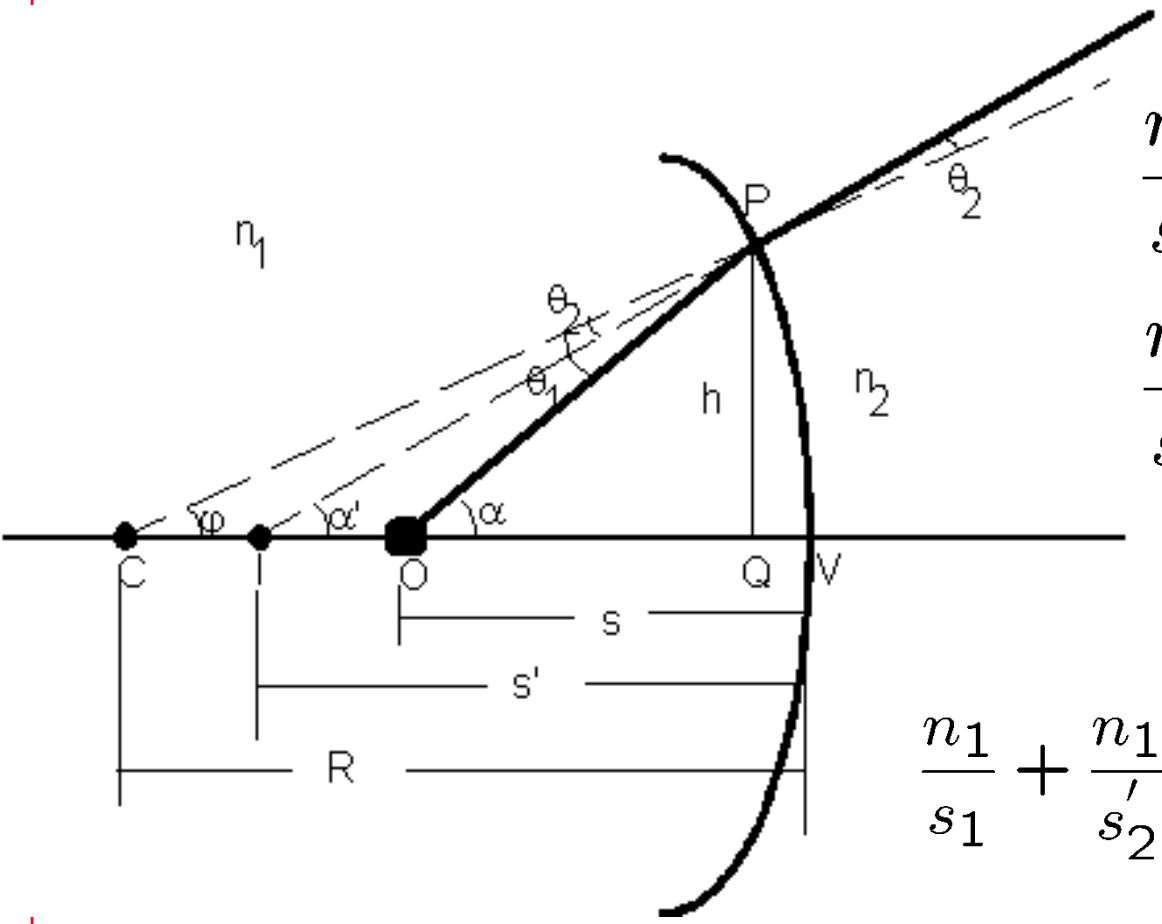
$$f' = \frac{n'r}{n'-n}$$

$$f = \frac{n.r}{n'-n}$$

Gaussian form: $\frac{f}{s} + \frac{f'}{s'} = 1$



Thin lens formula $n_1 \sin\theta_1 = n_2 \sin\theta_2$



$$\frac{n_1}{s_1} + \frac{n_2}{s_1'} = \frac{n_2 - n_1}{R_1}$$

$$\frac{n_2}{s_2} + \frac{n_1}{s_2'} = \frac{n_1 - n_2}{R_2}$$

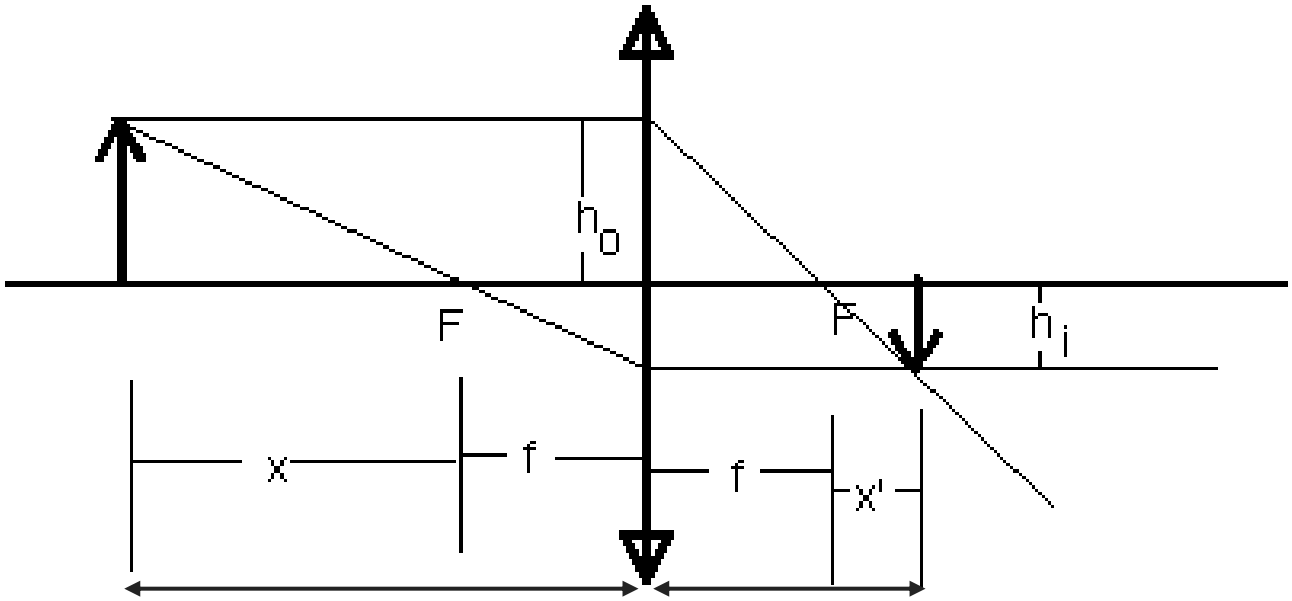
$s_2 = t - s'_1 \rightarrow s_2 = -s'_1$

$$\frac{n_1}{s_1} + \frac{n_1}{s_2'} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Lensmaker's equation:

$$\frac{1}{f} = \frac{n_2 - n_1}{n_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Thin lens formula $n_1 \sin\theta_1 = n_2 \sin\theta_2$



$$\frac{h_o}{p} = \frac{h_i}{q}$$

$$\frac{h_o}{p-f} = \frac{h_i}{f}$$

$$\frac{x}{f} = \frac{f}{x'}$$

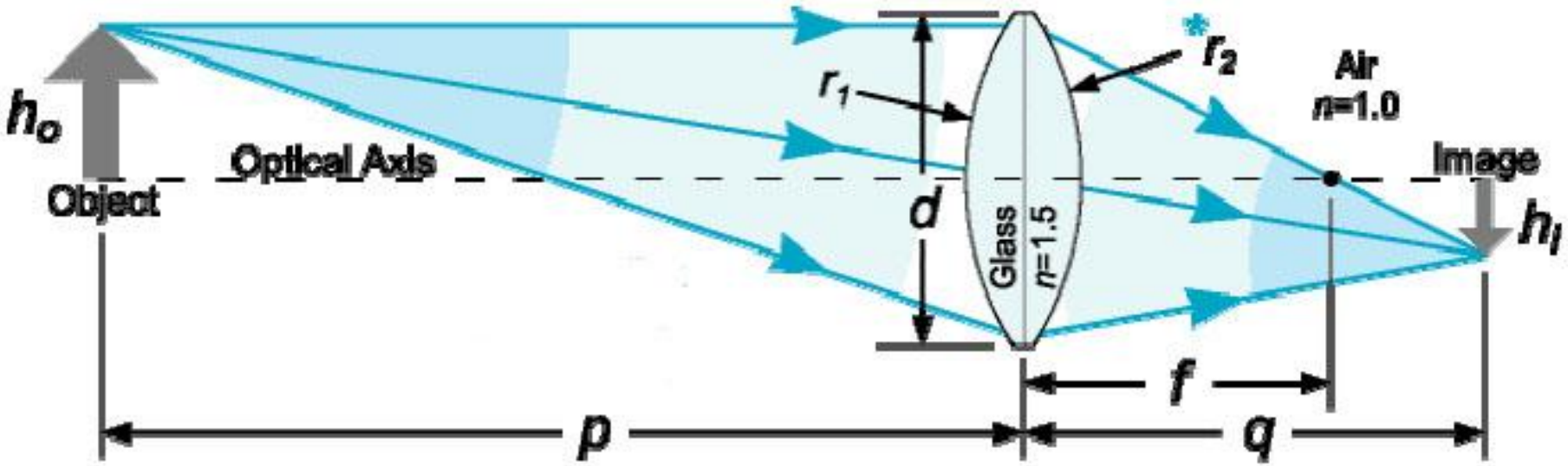
Newtonian Form:

$xx' = f^2$

Thin-Lens equantion:

$$\frac{1}{\mathbf{f}} = \frac{1}{p} + \frac{1}{q}$$

Thin lens formula



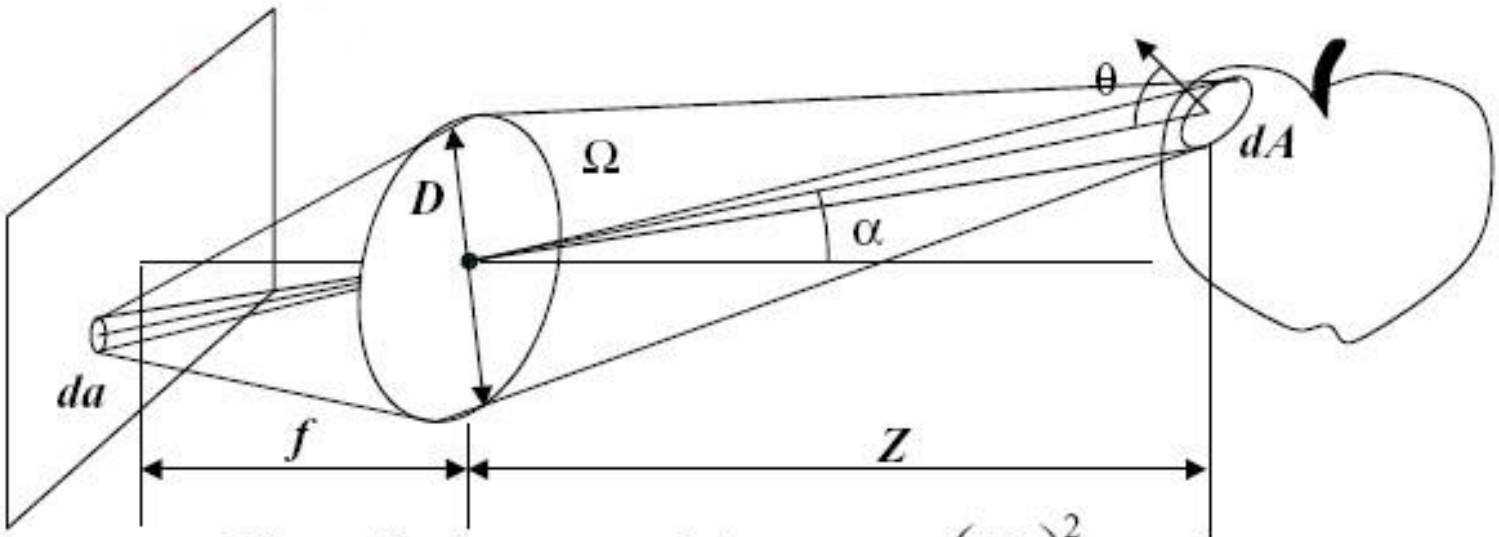
Lens Equation: $\frac{1}{f}=\frac{1}{p}+\frac{1}{q}$

Lens Maker's Equation: $\frac{1}{f}=(n-1)\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$

Magnification: $m=\frac{h_i}{h_o}=-\frac{q}{p}$

F-number: $f/\#=\frac{f}{d}$

Image intensity vs. illumination



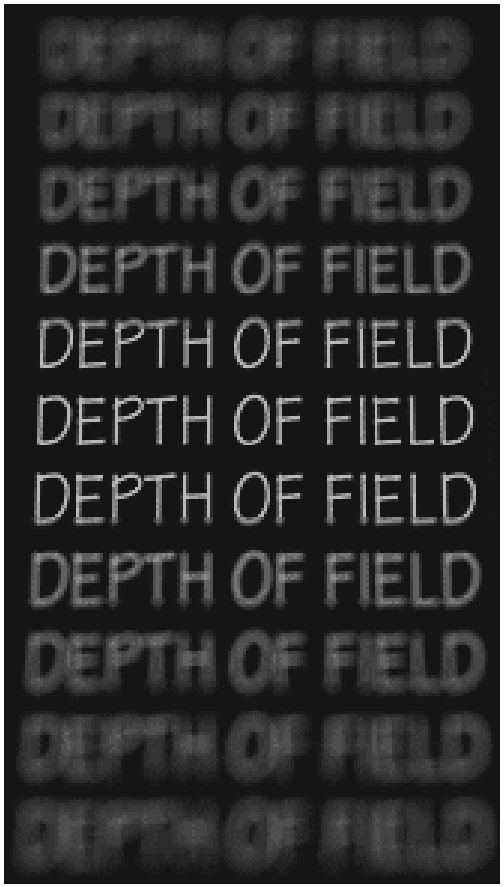
$$\frac{da\cos\alpha}{(f/\cos\alpha)^2}=\frac{dA\cos\theta}{(Z/\cos\alpha)^2}\Rightarrow\frac{dA}{da}=\frac{\cos\alpha}{\cos\theta}\left(\frac{Z}{f}\right)^2$$

$$dP=LdA\,\Omega\cos\theta\Rightarrow dP=LdA\frac{\pi}{4}\left(\frac{D}{Z}\right)^2\cos^3\alpha\cos\theta$$

$$E=\frac{dP}{da}=L\frac{dA}{da}\frac{\pi}{4}\left(\frac{D}{Z}\right)^2\cos^3\alpha\ \cos\theta\Rightarrow E=\frac{\pi}{4}\left(\frac{D}{f}\right)^2\cos^4\alpha\ \ L$$

Depth of Field

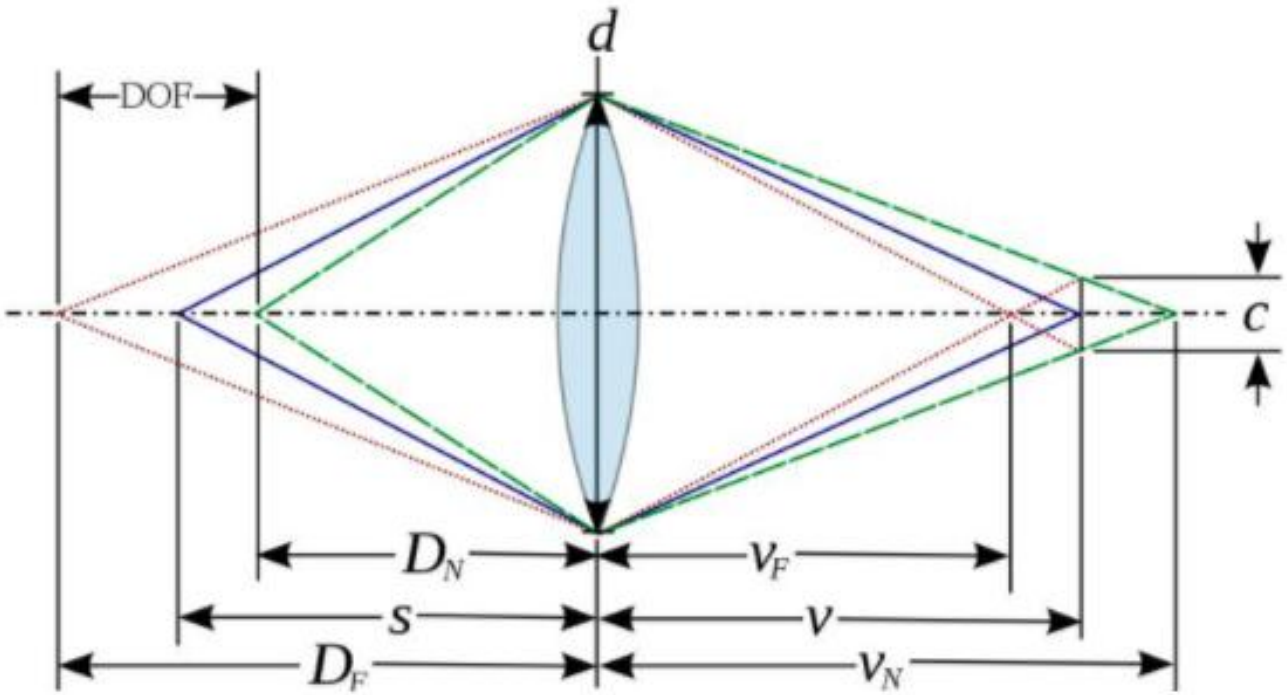




http://www.cambridgeincolour.com/tutorials/depth-of-field.htm

$$D_N = \frac{sf^2}{f^2 - cNf + cNs}$$

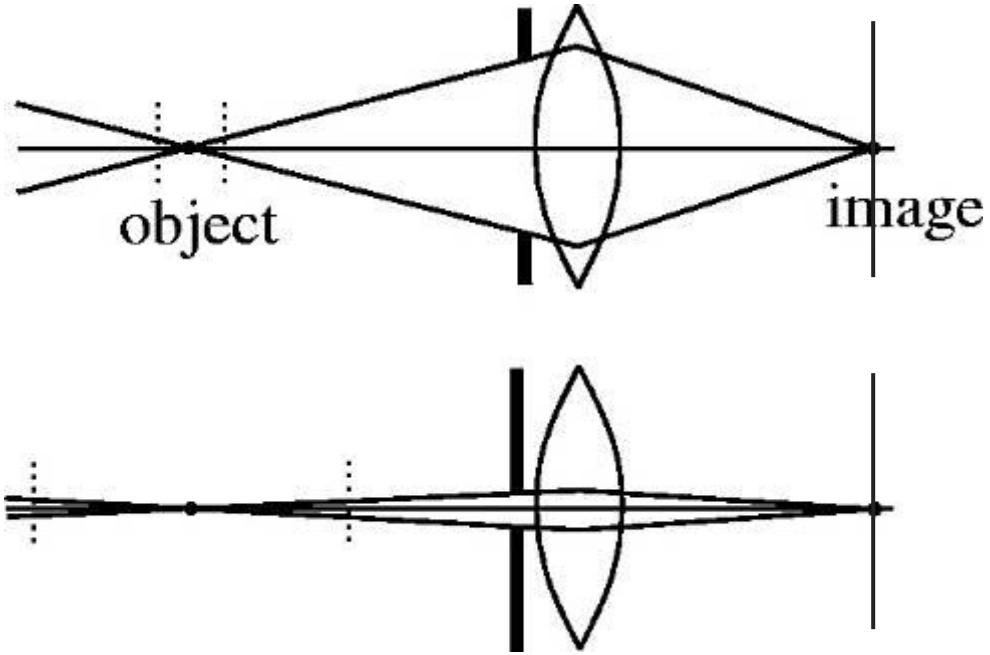
$$D_F = \frac{sf^2}{f^2 + cNf - cNs}$$



其中 DN 为后物体成像，DF 为前物体成像，两者之差即为景深。

s 为物距，f 为镜头焦距，N 为镜头 f 值，C 是弥散圆。

How can we control the depth of field?



Changing the aperture size affects depth of field

- A smaller aperture increases the range in which the object is approximately in focus
- But small aperture reduces amount of light – need to increase exposure

Slide by A. Efros

Varying the aperture



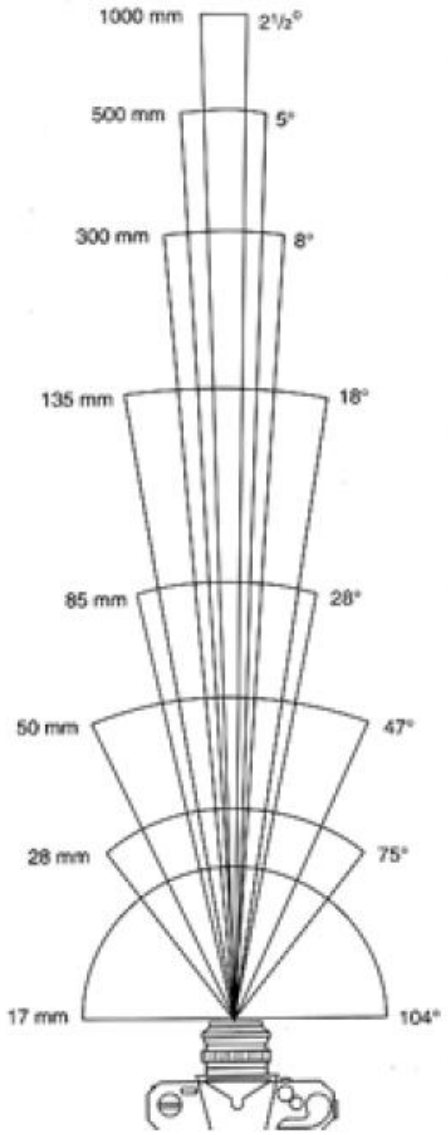


Large aperture = small DOF

Small aperture = large DOF

Slide by A. Efros

Field of View





17mm



28mm



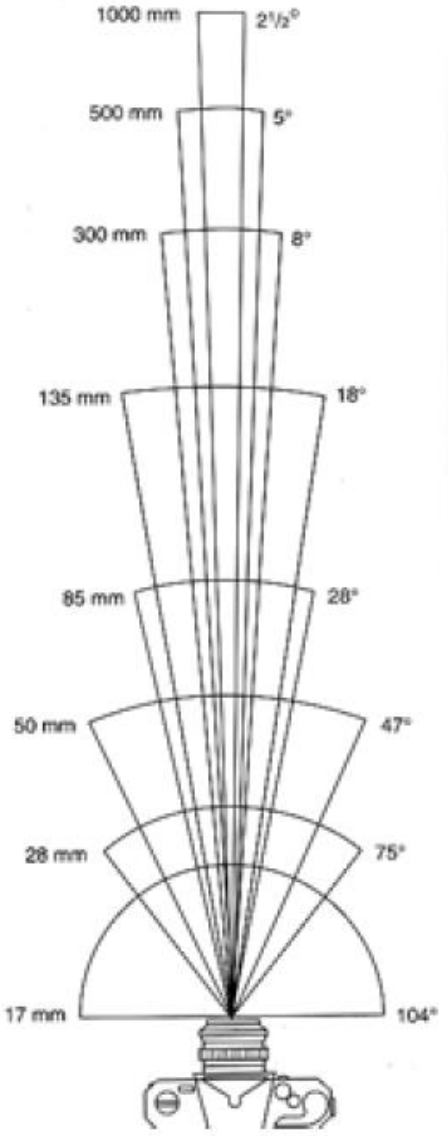
50mm



85mm

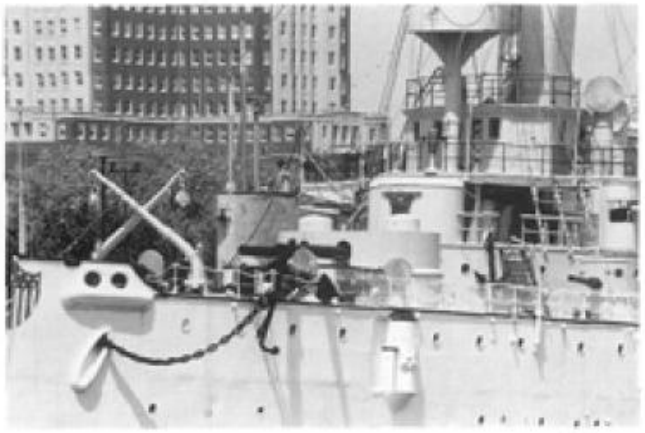
Slide by A. Efros

Field of View





135mm



300mm

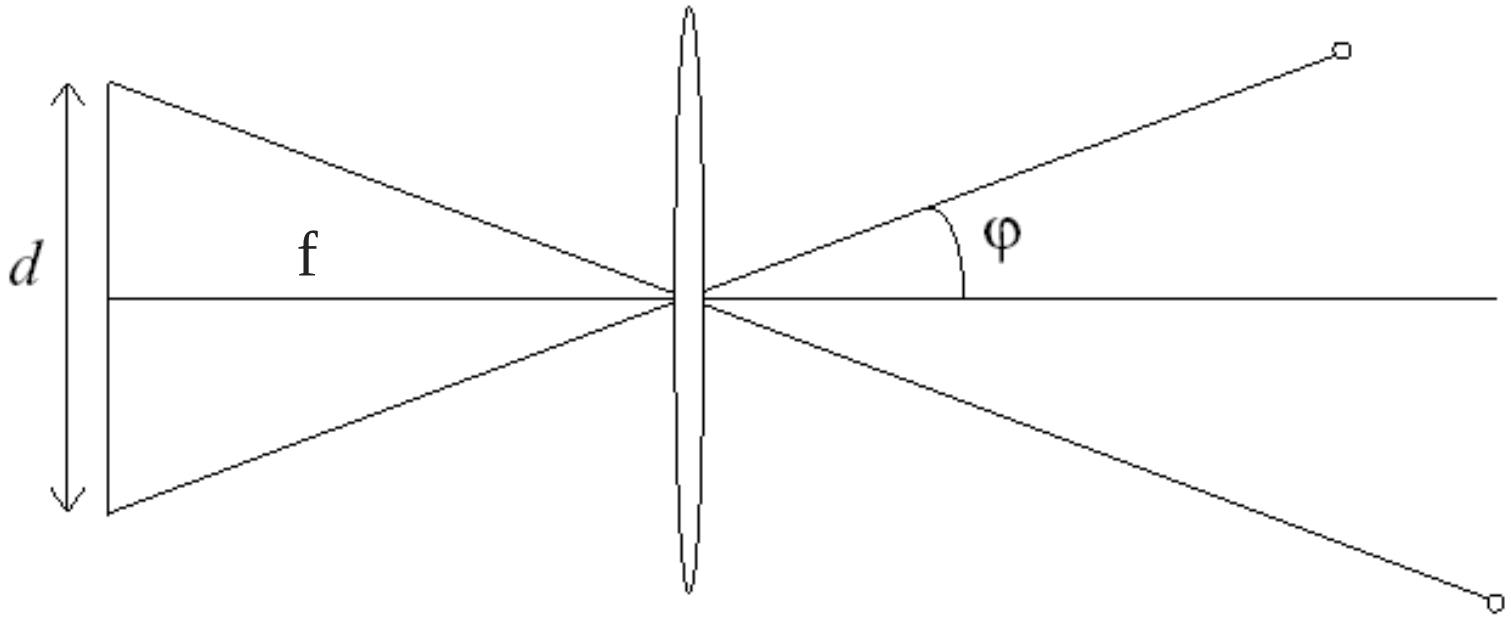




What does FOV depend on?

Slide by A. Efros

Field of View



FOV depends on focal length and size of the camera retina

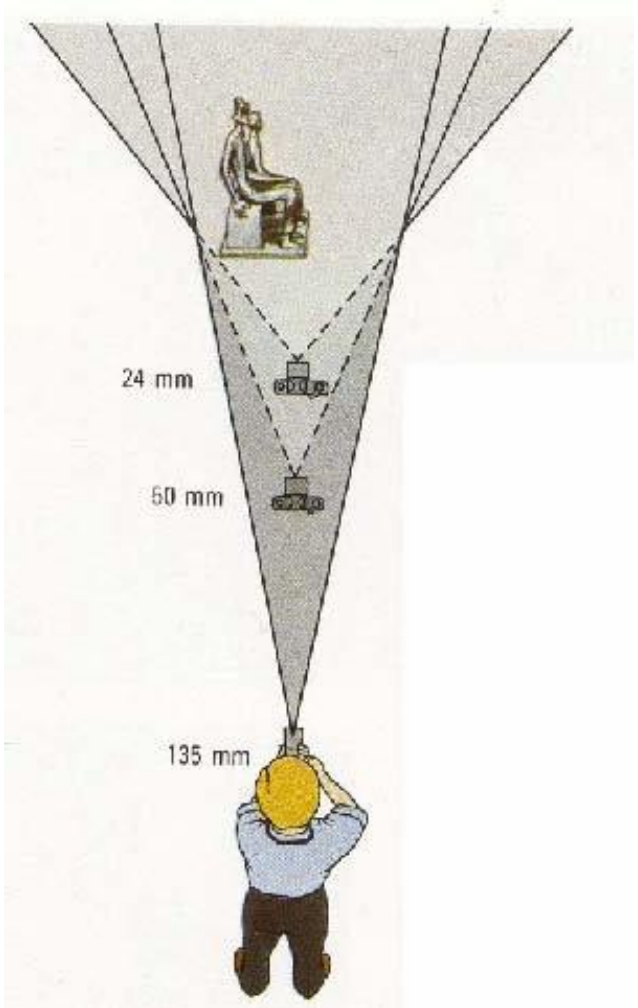
$$\varphi = \tan^{-1}\left(\frac{d}{2f}\right)$$

Smaller FOV = larger Focal Length

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Slide by A. Efros

Field of View / Focal Length





Large FOV, small f
Camera close to car



Small FOV, large f
Camera far from the car

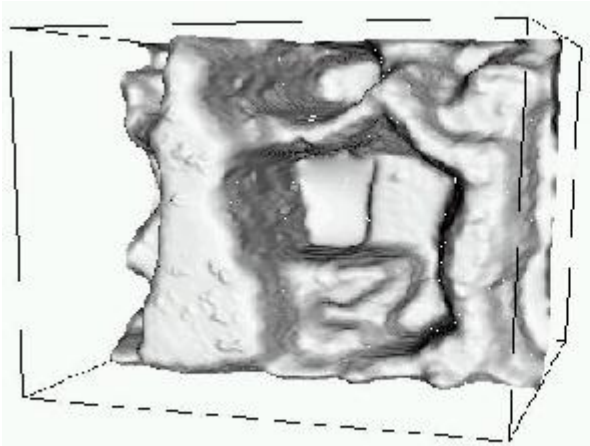
Sources: A. Efros, F. Durand

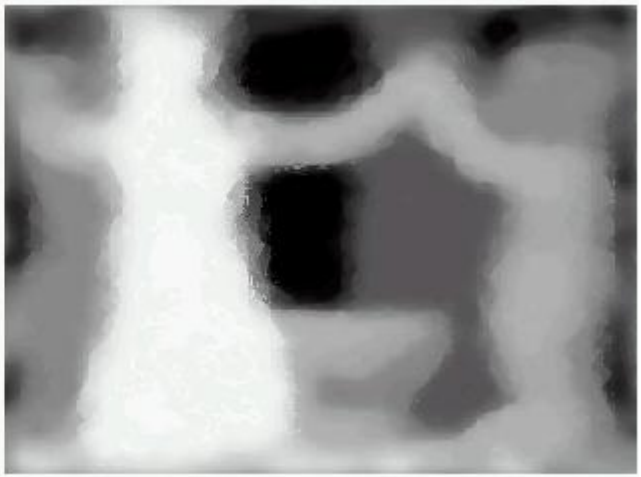
Depth from focus





Images from same point of view, different camera parameters

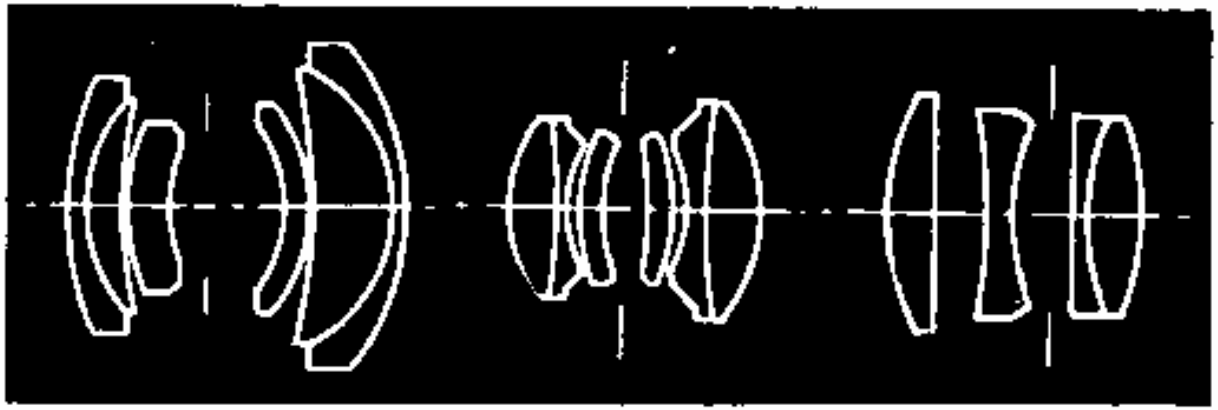




3d shape / depth estimates

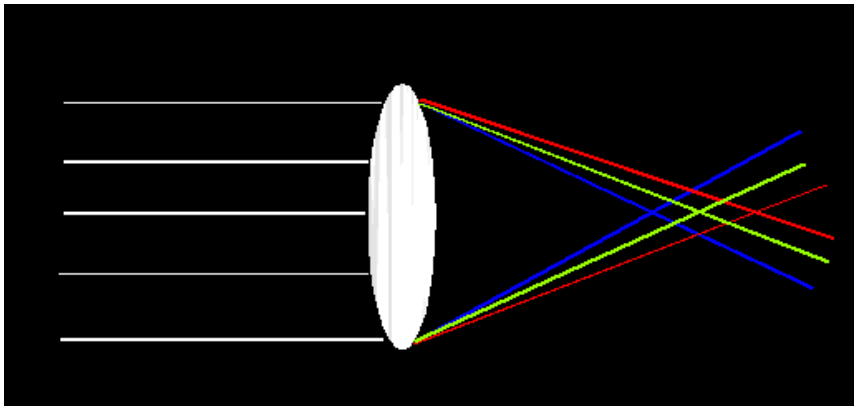
[figs from H. Jin and P. Favaro, 2002]

Real lenses



Lens Flaws: Chromatic Aberration

Lens has different refractive indices for different wavelengths: causes color fringing



Near Lens Center



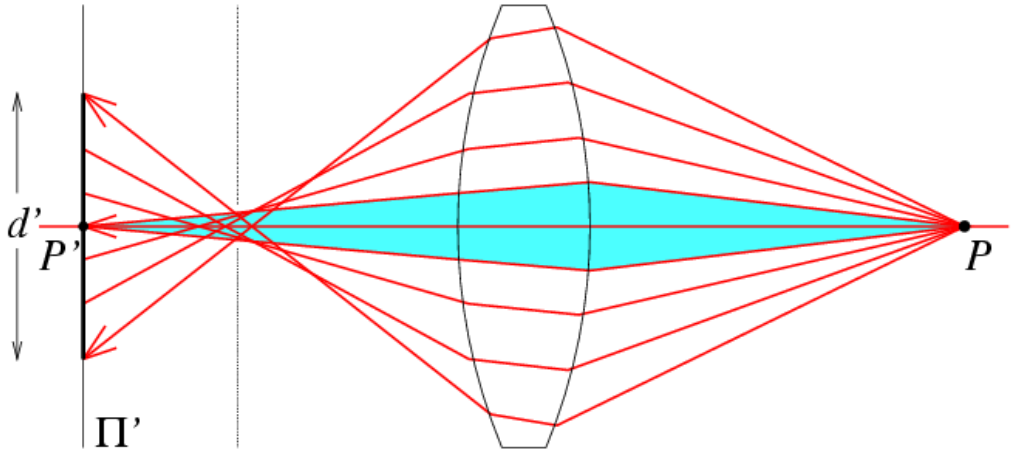
Near Lens Outer Edge



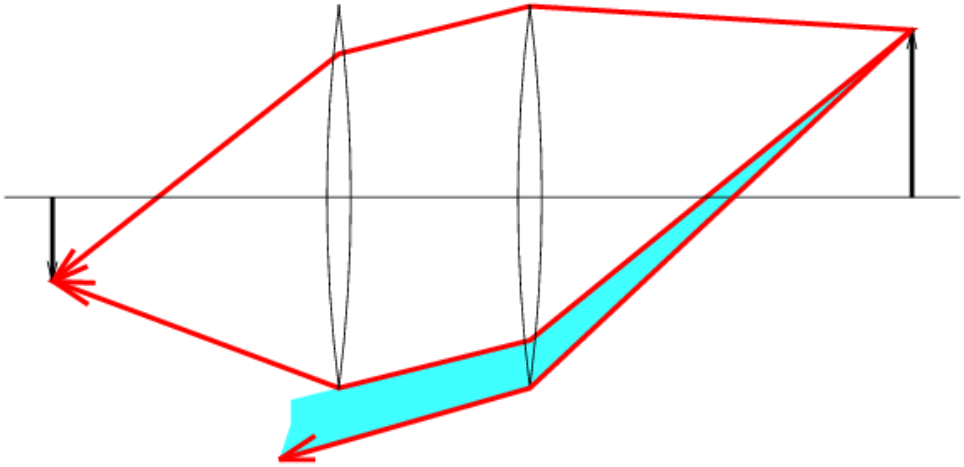
Lens flaws: Spherical aberration

Spherical lenses don’t focus light perfectly

Rays farther from the optical axis focus closer



Lens flaws: Vignetting



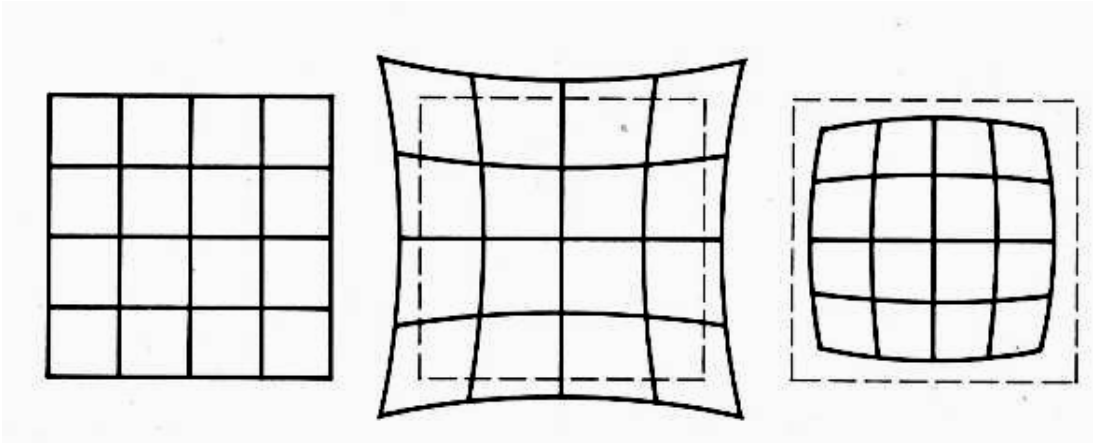




Radial Distortion

Caused by imperfect lenses

Deviations are most noticeable near the edge of the lens

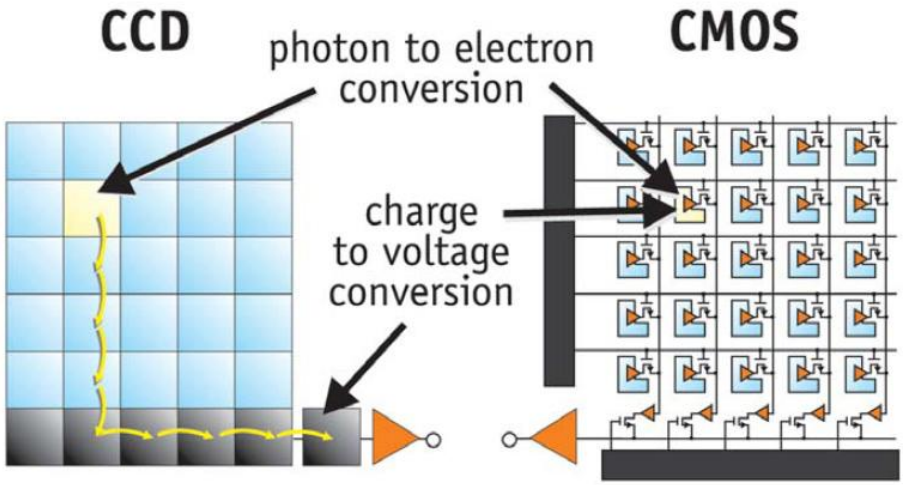


No distortion Pin cushion Barrel



Digital camera





CCDs move photogenerated charge from pixel to pixel and convert it to voltage at an output node. CMOS imagers convert charge to voltage inside each pixel.

A digital camera replaces film with a sensor array

Each cell in the array is light-sensitive diode that converts photons to electrons

Two common types

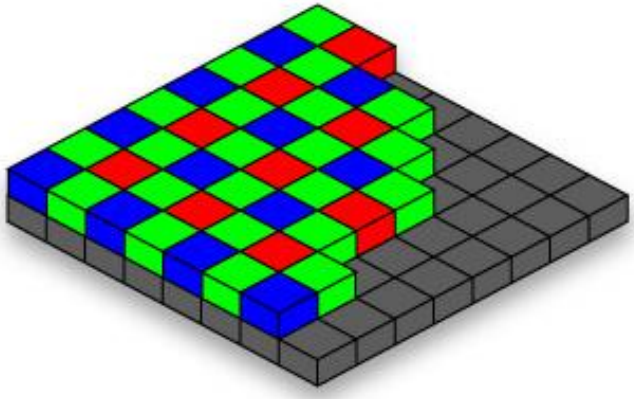
Charge Coupled Device (CCD)

Complementary metal oxide semiconductor (CMOS)

http://electronics.howstuffworks.com/digital-camera.htm

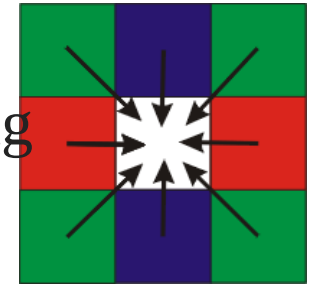
Color filter arrays

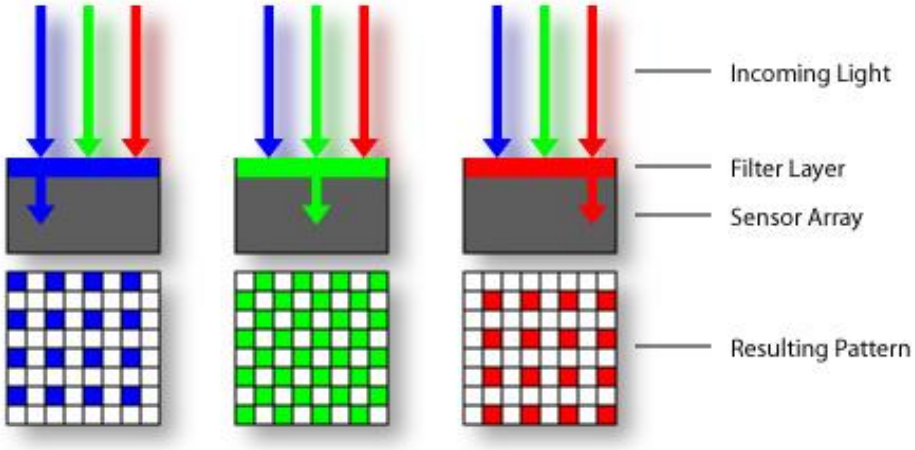
Bayer grid



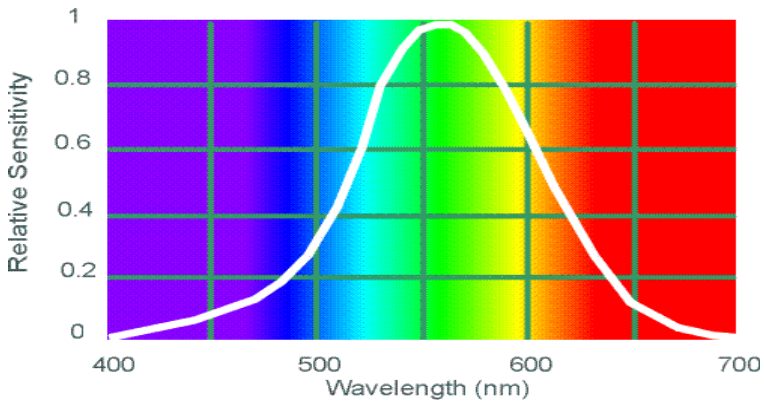
Demosaicing:

Estimation of missing components from neighboring values





Why more green?



Human Luminance Sensitivity Function

Source: Steve Seitz

Misc. digital camera artifacts

Noise

low light is where you most notice noise
light sensitivity (ISO) / noise tradeoff
stuck pixels



In-camera processing

oversharpening can produce halos



Compression

JPEG artifacts, blocking

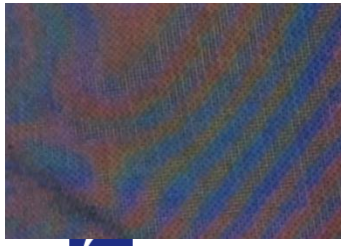


Blooming

CCD charge overflowing into neighboring pixels

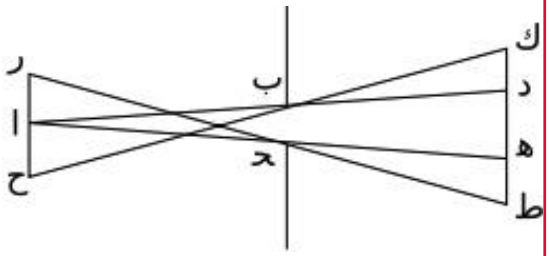
Color artifacts

Color moire
Purple fringing from microlenses



Historic milestones

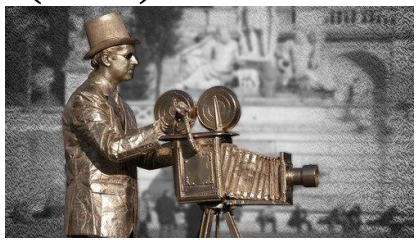
- **Pinhole model:** Mozi (470-390 BCE), Aristotle (384-322 BCE)
- **Principles of optics (including lenses):** Alhacen (965-1039 CE)
- **Camera obscura:** Leonardo da Vinci (1452-1519), Johann Zahn (1631-1707)
- **First photo:** Joseph Nicephore Niepce (1822)
- **Daguerréotypes** (1839)
- **Photographic film** (Eastman, 1889)
- **Cinema** (Lumière Brothers, 1895)
- **Color Photography** (Lumière Brothers, 1908)
- **Television** (Baird, Farnsworth, Zworykin, 1920s)
- **First consumer camera with CCD** Sony Mavica (1981)
- **First fully digital camera:** Kodak DCS100 (1990)



Alhacen's notes



Niepce, "La Table Servie," 1822





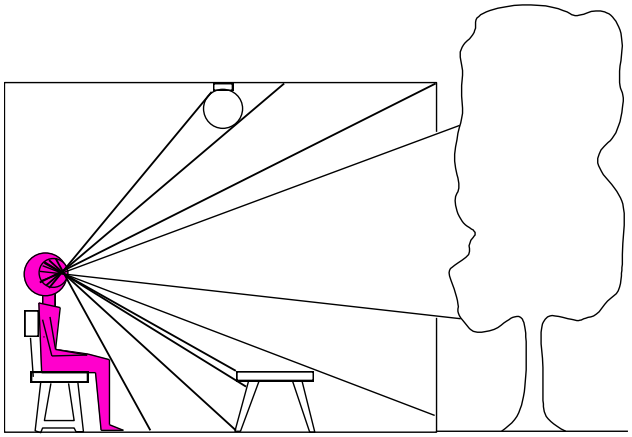
Old television camera

https://en.wikipedia.org/wiki/History_of_photography

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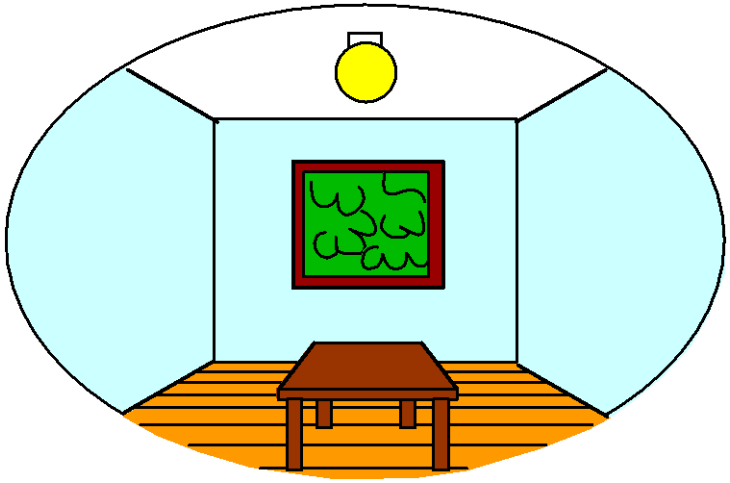
Dimensionality reduction: from 3D to 2D

3D world



Point of observation

2D image



What is preserved?

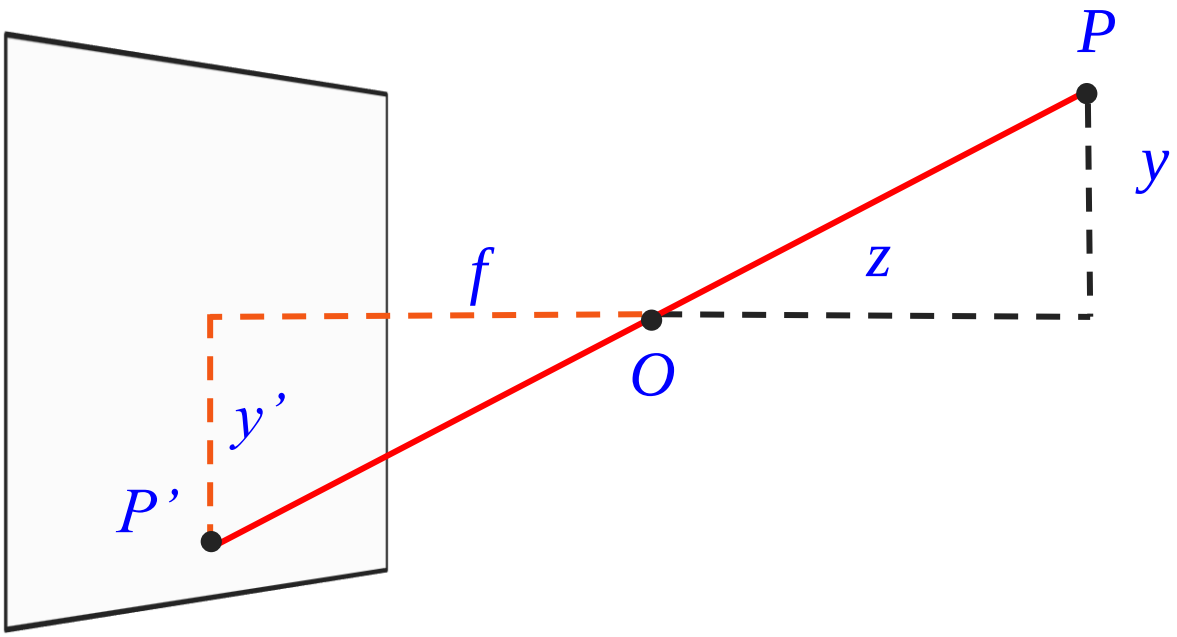
- Straight lines, incidence

What have we lost?

- Angles, lengths

Figures © Stephen E. Palmer, 2002

Pinhole projection model



Projection equations

- Derived using similar triangles

$$(x, y, z) \rightarrow (f\frac{x}{z}, f\frac{y}{z})$$

Note: instead of dealing with an image that is upside down, most of the time we will pretend that the image plane is in front of the camera center.

Projection properties

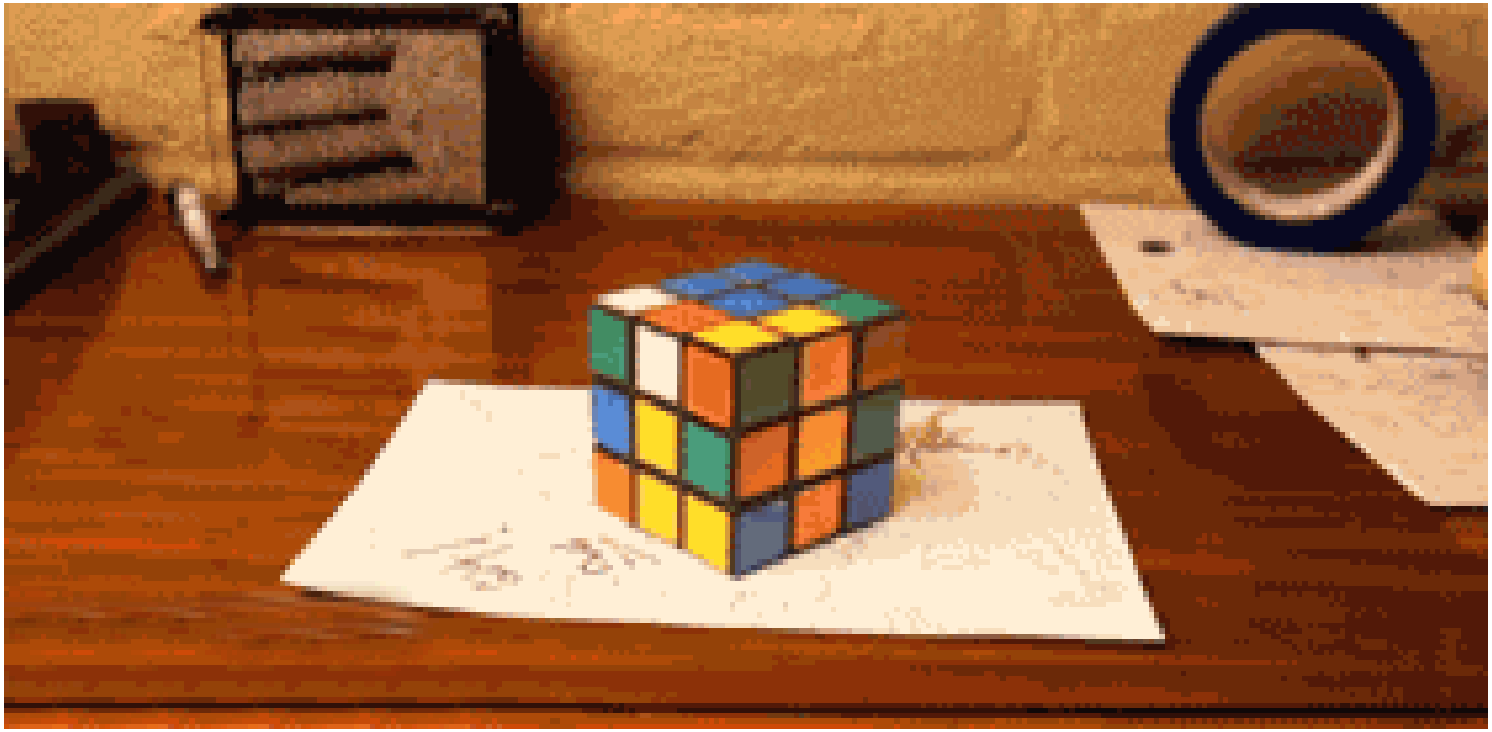
- Many-to-one: any points along same *visual ray* map to same point in image
- Points → points

  But projection of points on *focal plane* is undefined
- Lines → lines (collinearity is preserved)

  But lines through focal point (visual rays) project to a point
- Planes → planes (or half-planes)

  But planes through focal point project to lines

Projection can be tricky…



Projection can be tricky…



Projection can be tricky…



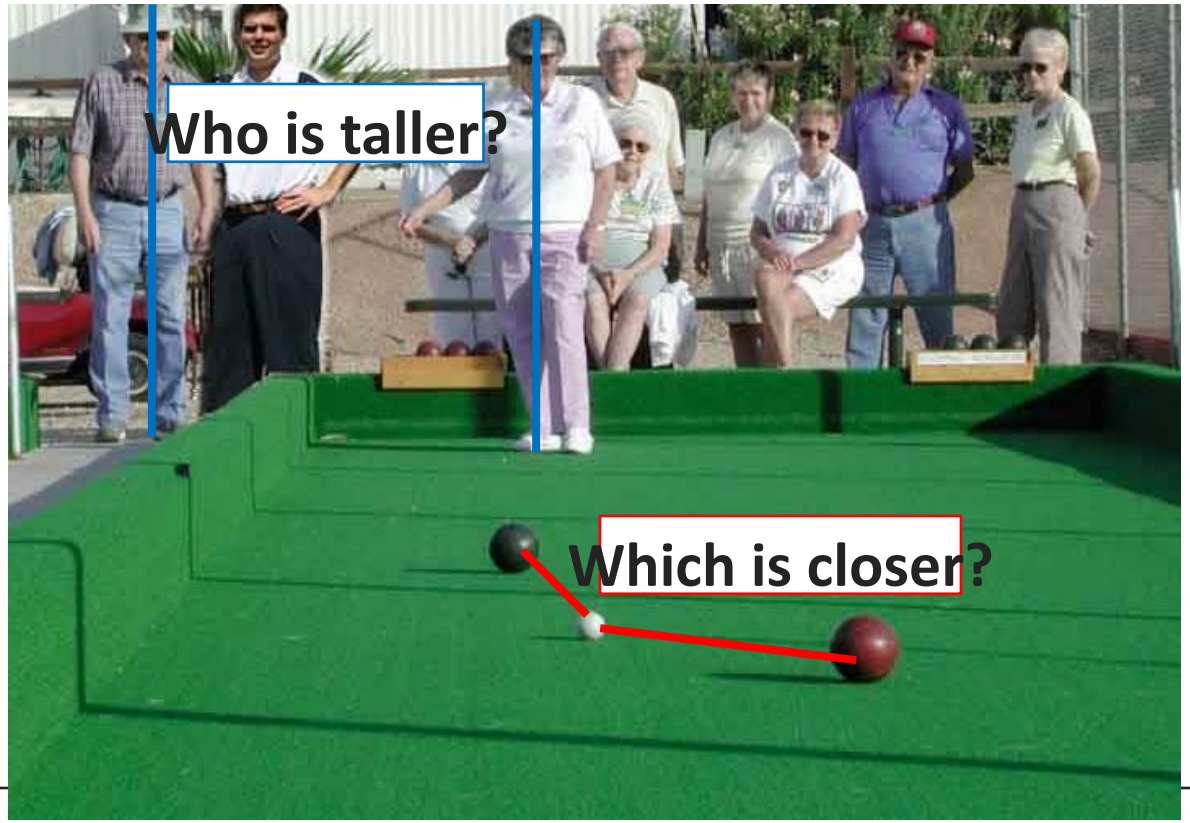
Projection can be tricky…



Projective Geometry

What is lost?

Length



Length is not preserved

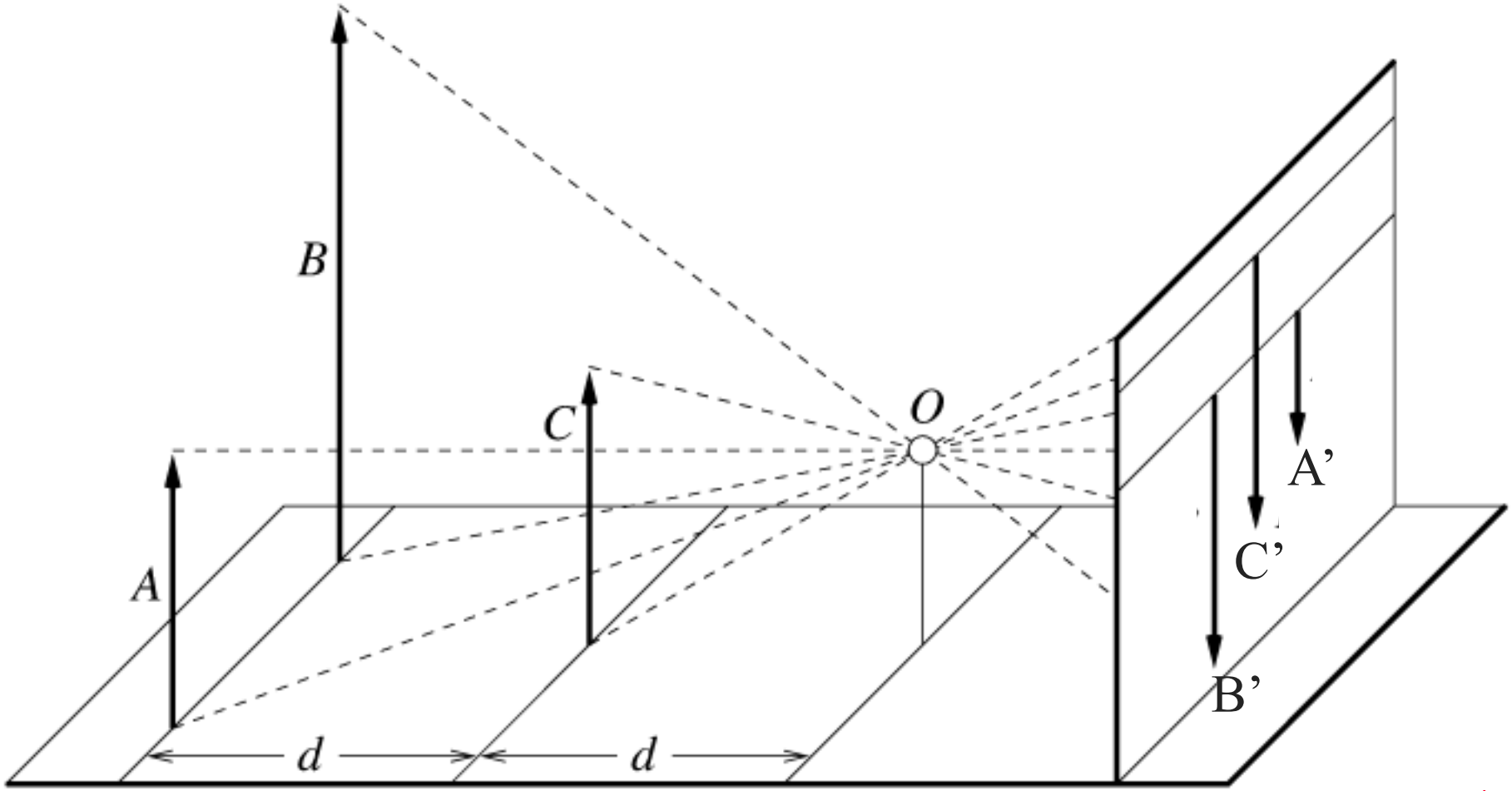


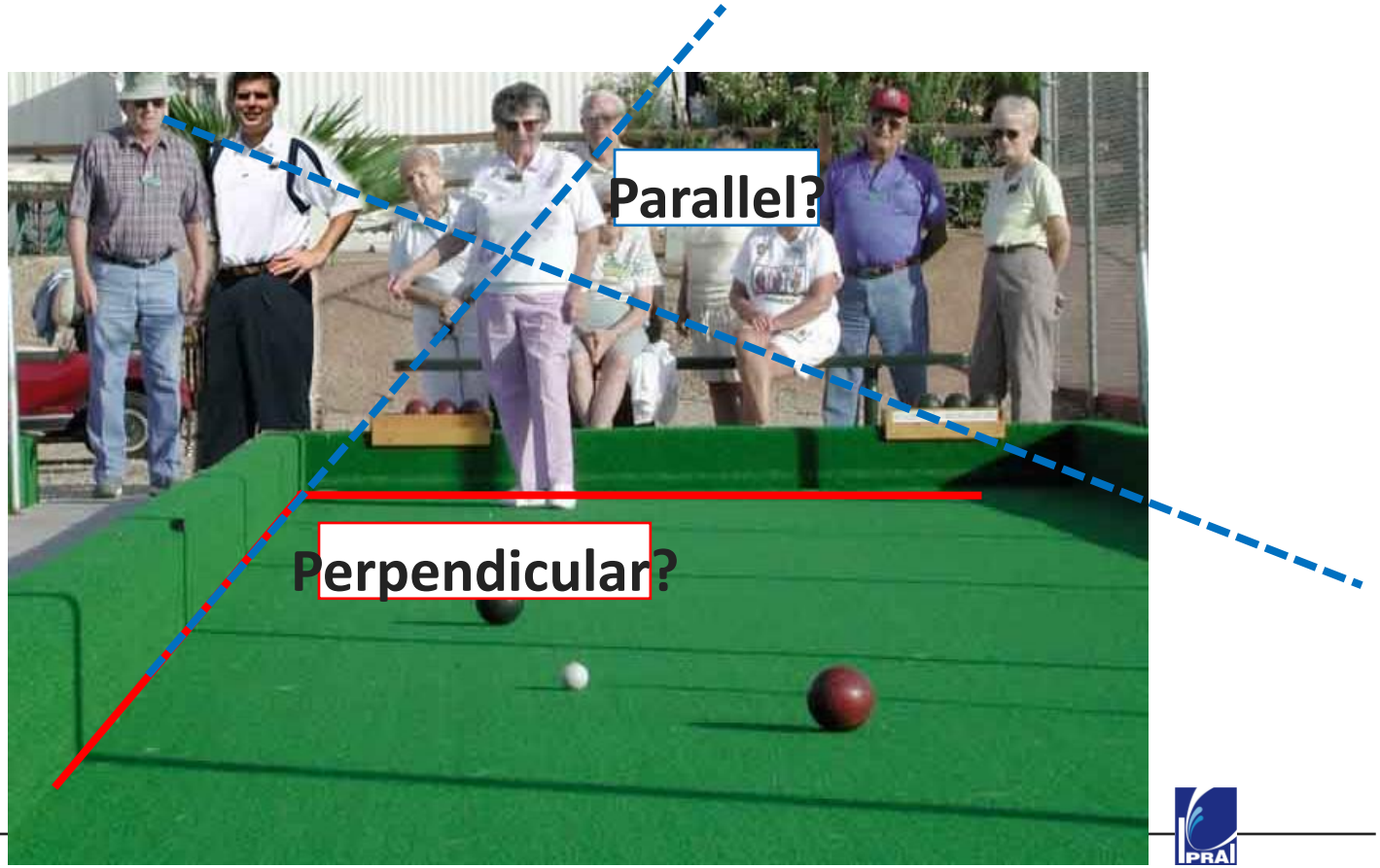
Figure by David Forsyth

Projective Geometry

What is lost?

Length

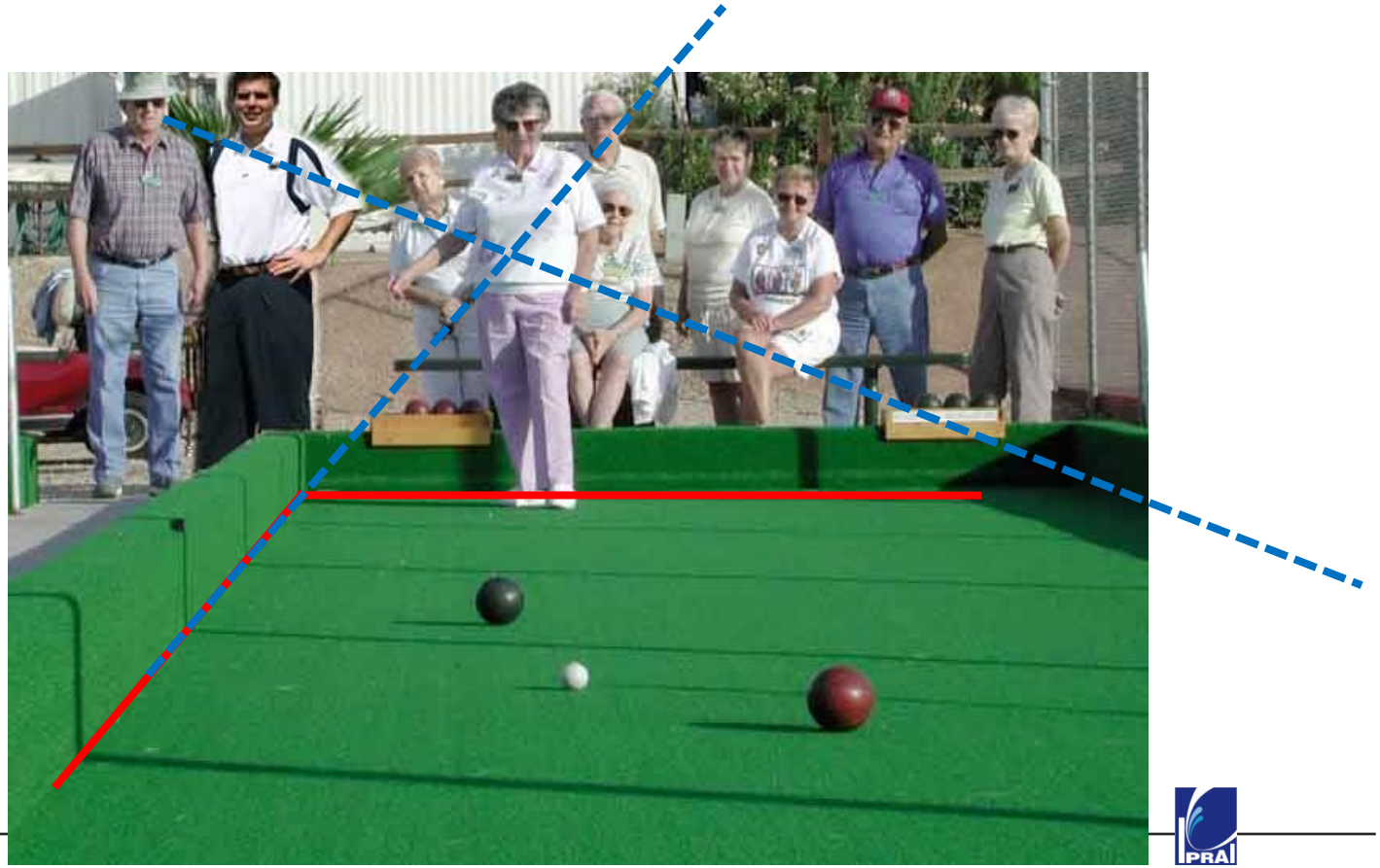
Angles



Projective Geometry

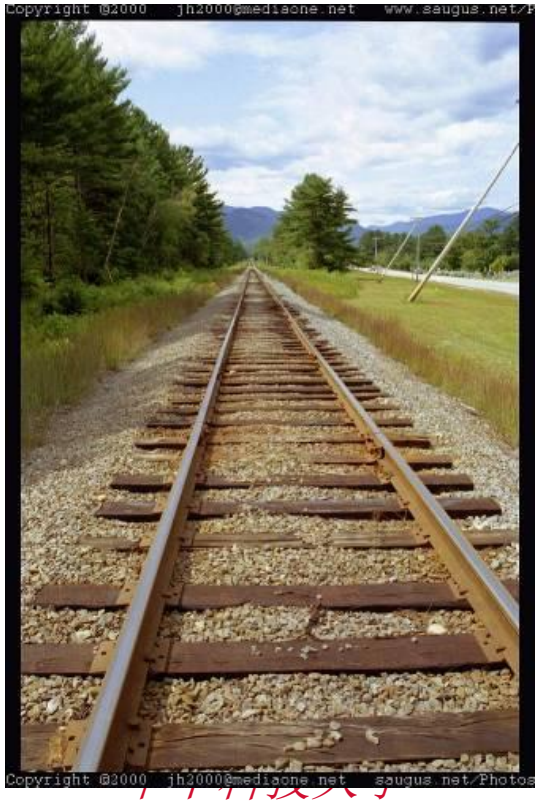
What is preserved?

Straight lines are still straight



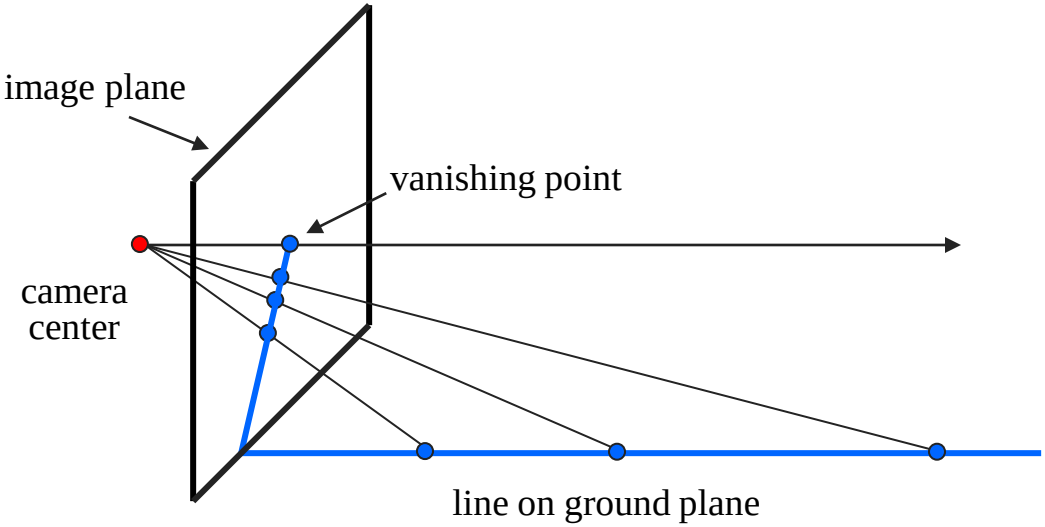
Vanishing points

- Each direction in space has its own vanishing point
  - All lines going in that direction converge at that point
  - Exception: directions parallel to the image plane

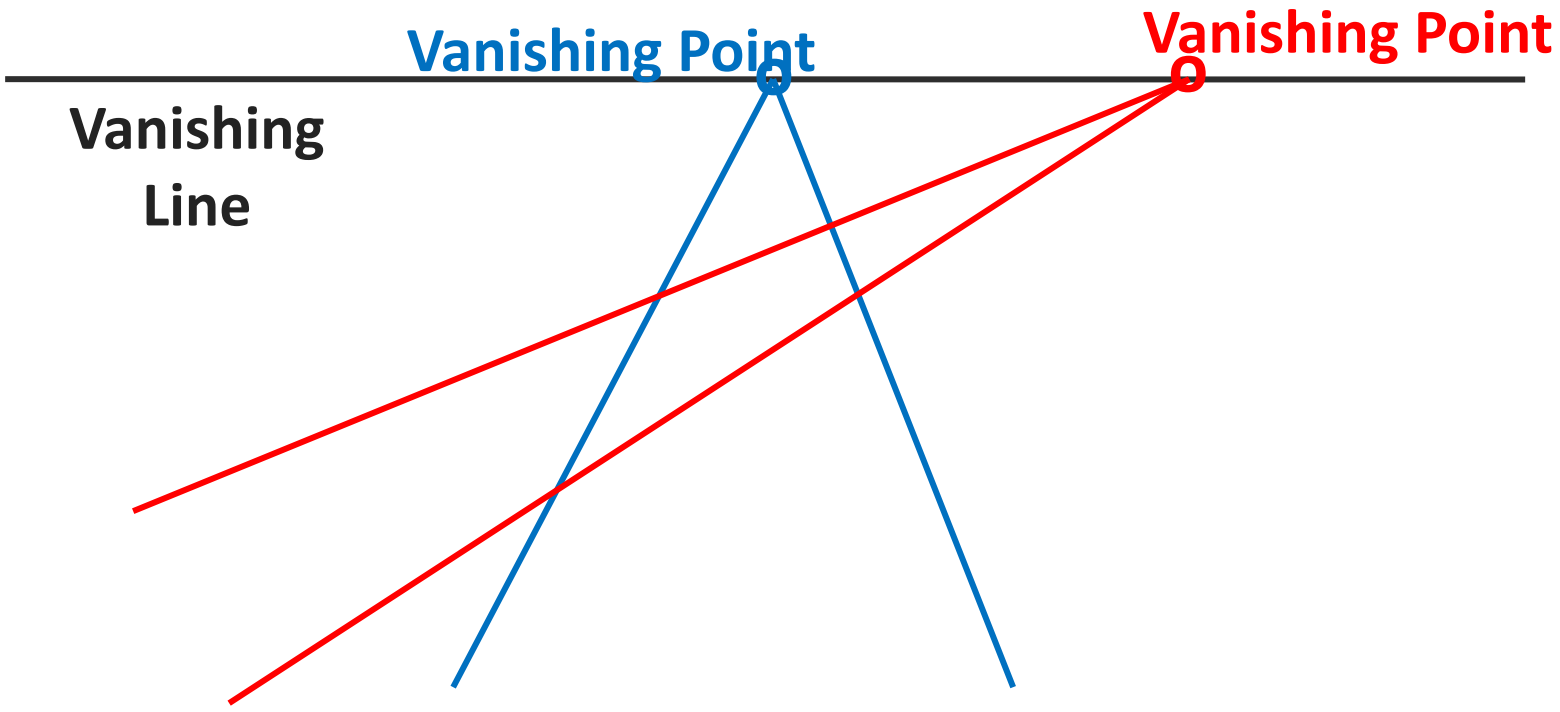


Vanishing points

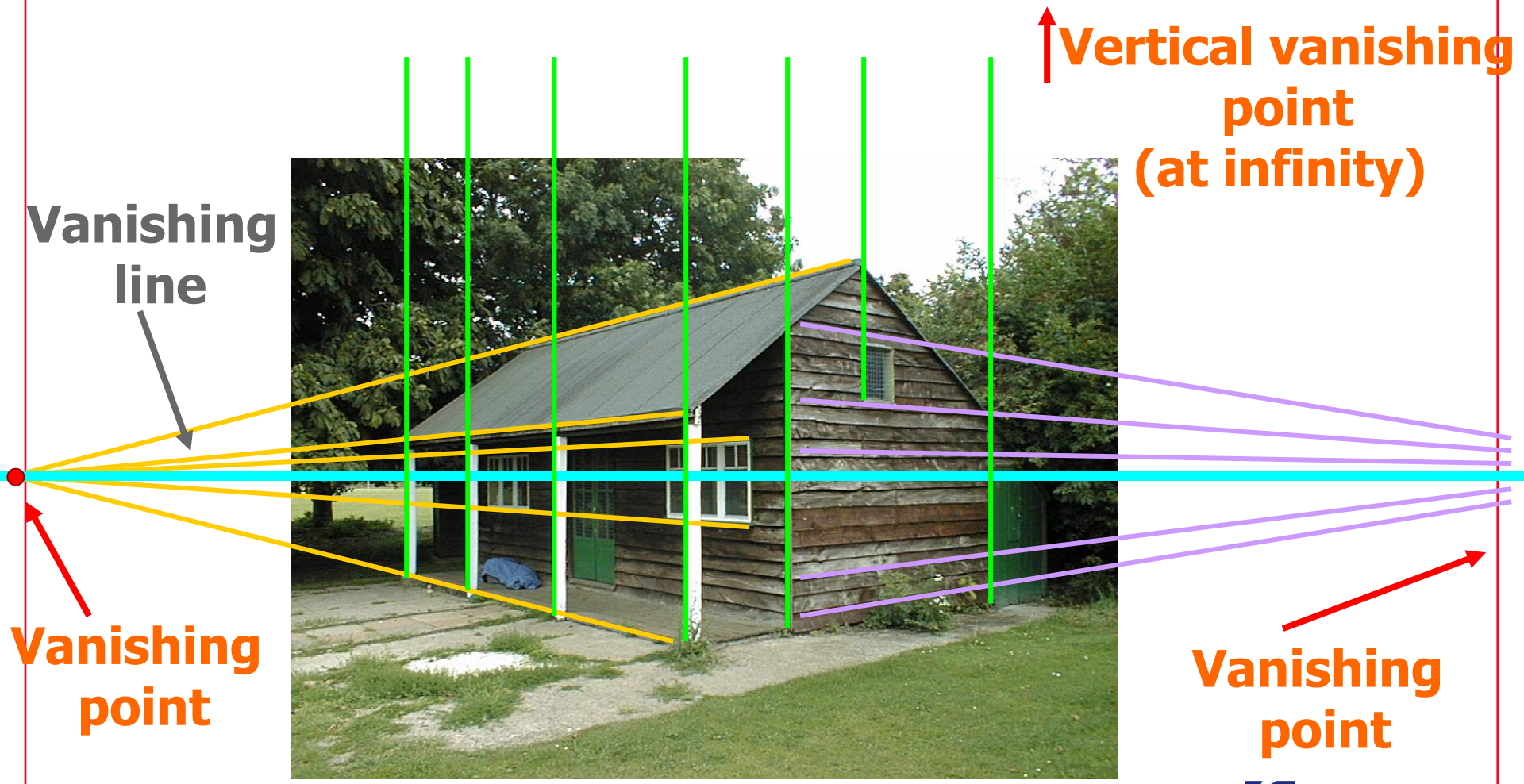
- Each direction in space has its own vanishing point
  - All lines going in that direction converge at that point
  - Exception: directions parallel to the image plane
- How do we construct the vanishing point of a line?
  - What about the vanishing line of a plane?



Vanishing points and lines



Vanishing points and lines



Slide from Efros, Photo from Criminisi

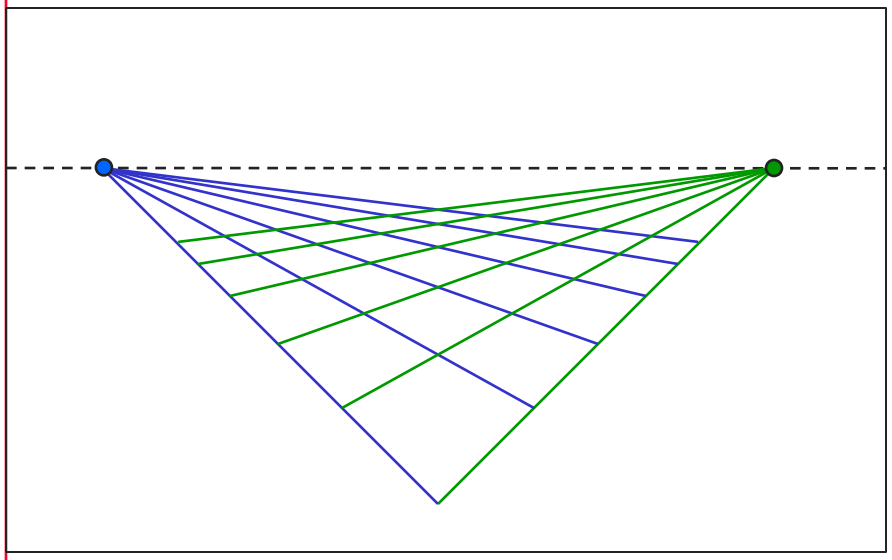
Vanishing lines of planes

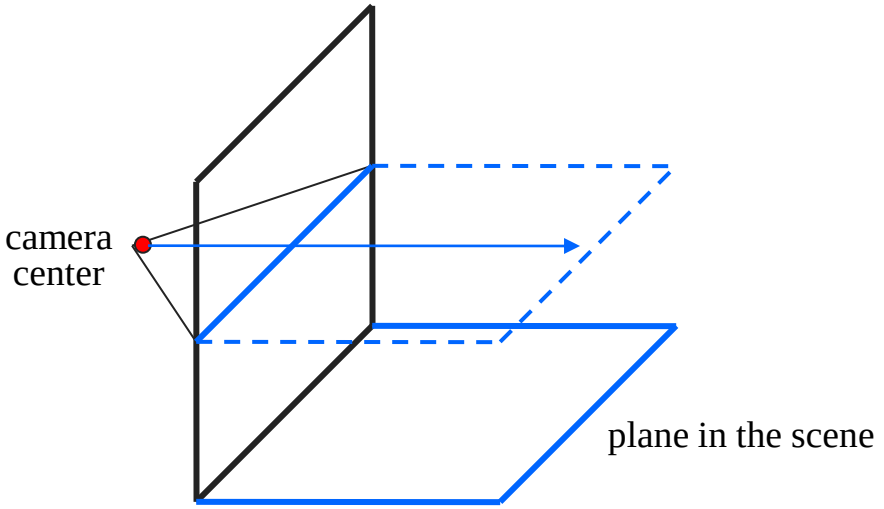


How do we construct the vanishing line of a plane?

Image source

Vanishing lines of planes





- *Horizon*: vanishing line of the ground plane
  - All points at the same height as the camera project to the horizon
  - Points higher (resp. lower) than the camera project above (resp. below) the horizon
  - Provides way of comparing height of objects

Slide by Steve Seitz

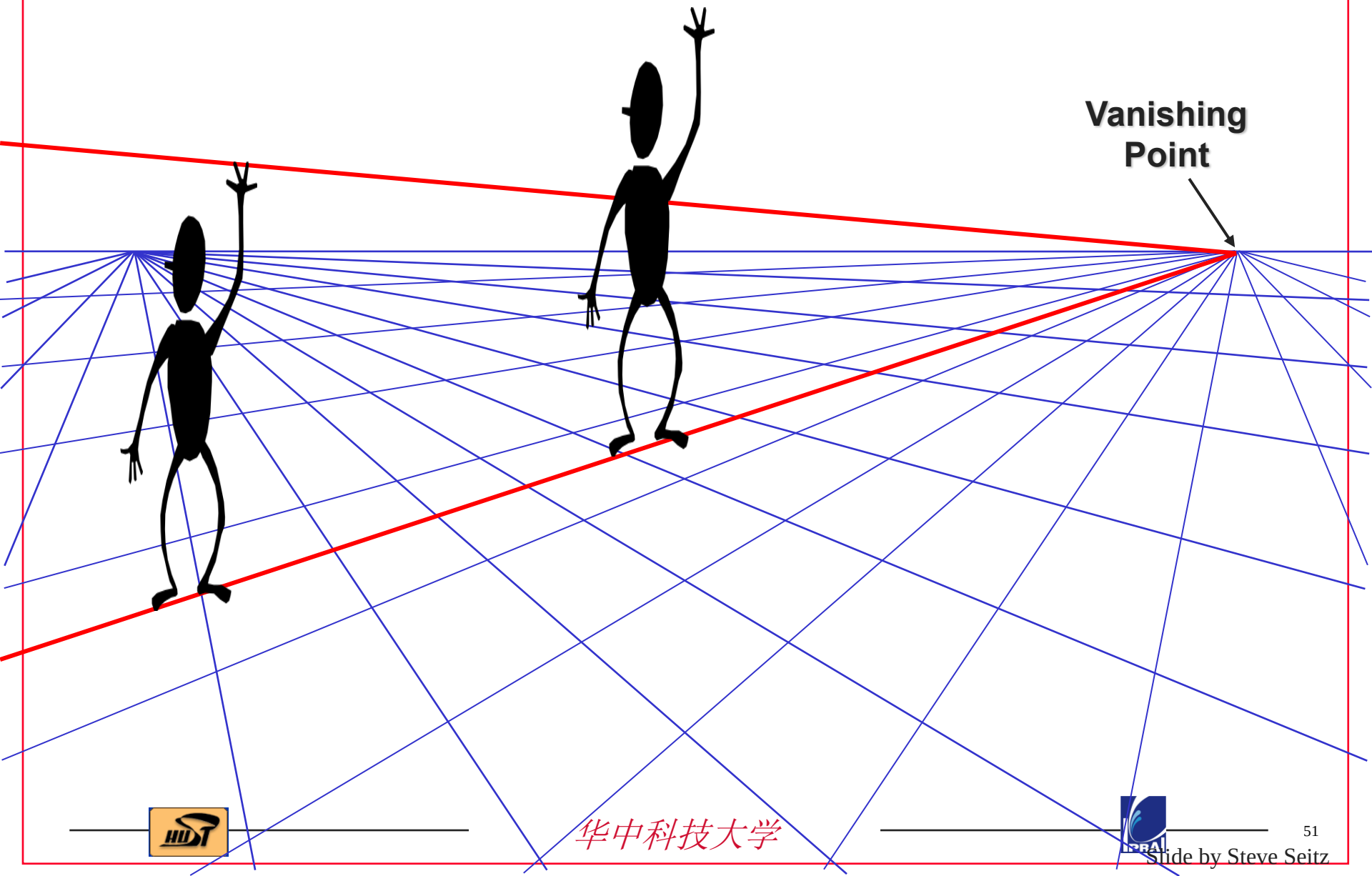
Vanishing lines of planes



Is the parachutist above or below the camera?

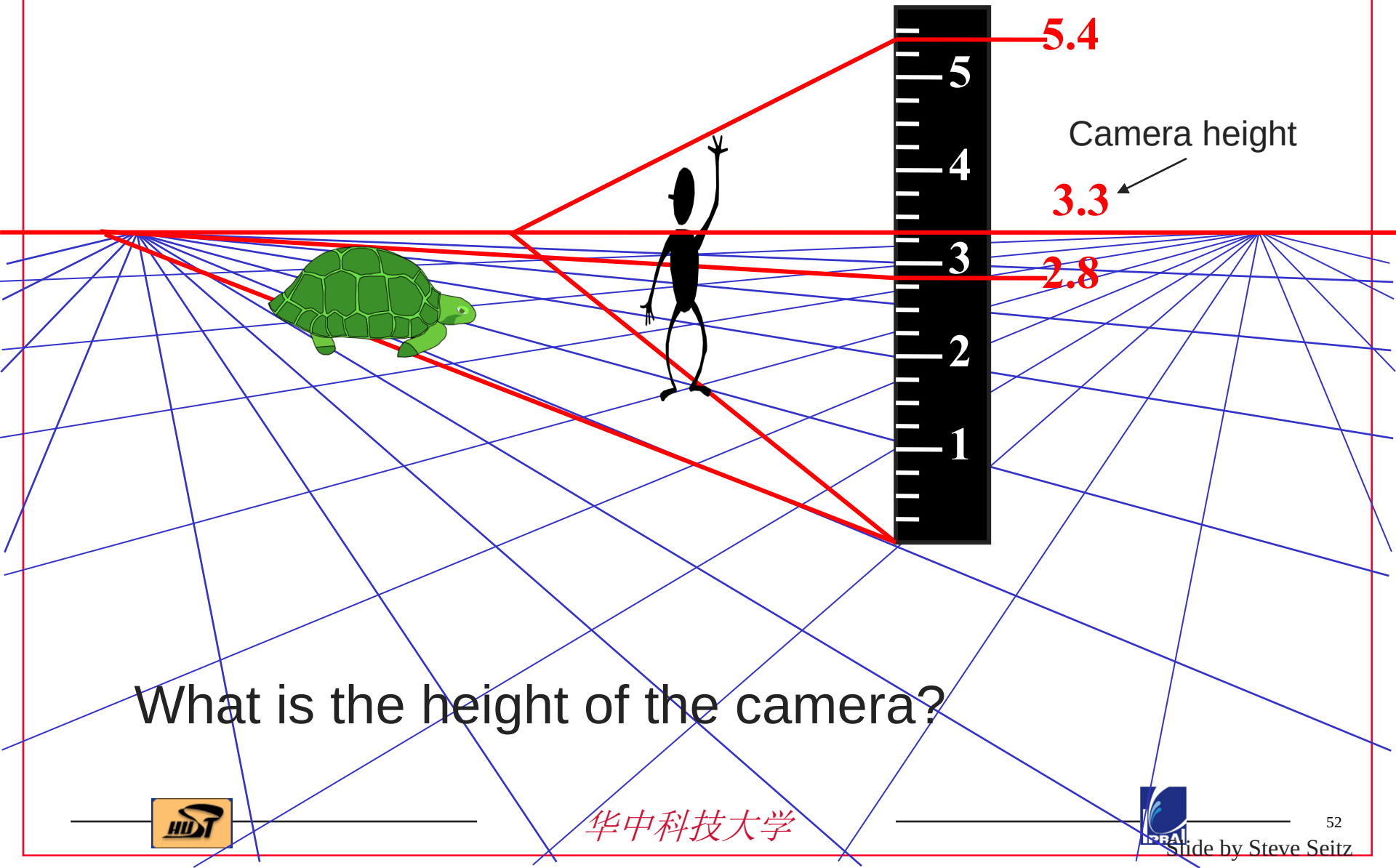
Image source: S. Seitz

Comparing heights



Slide by Steve Seitz

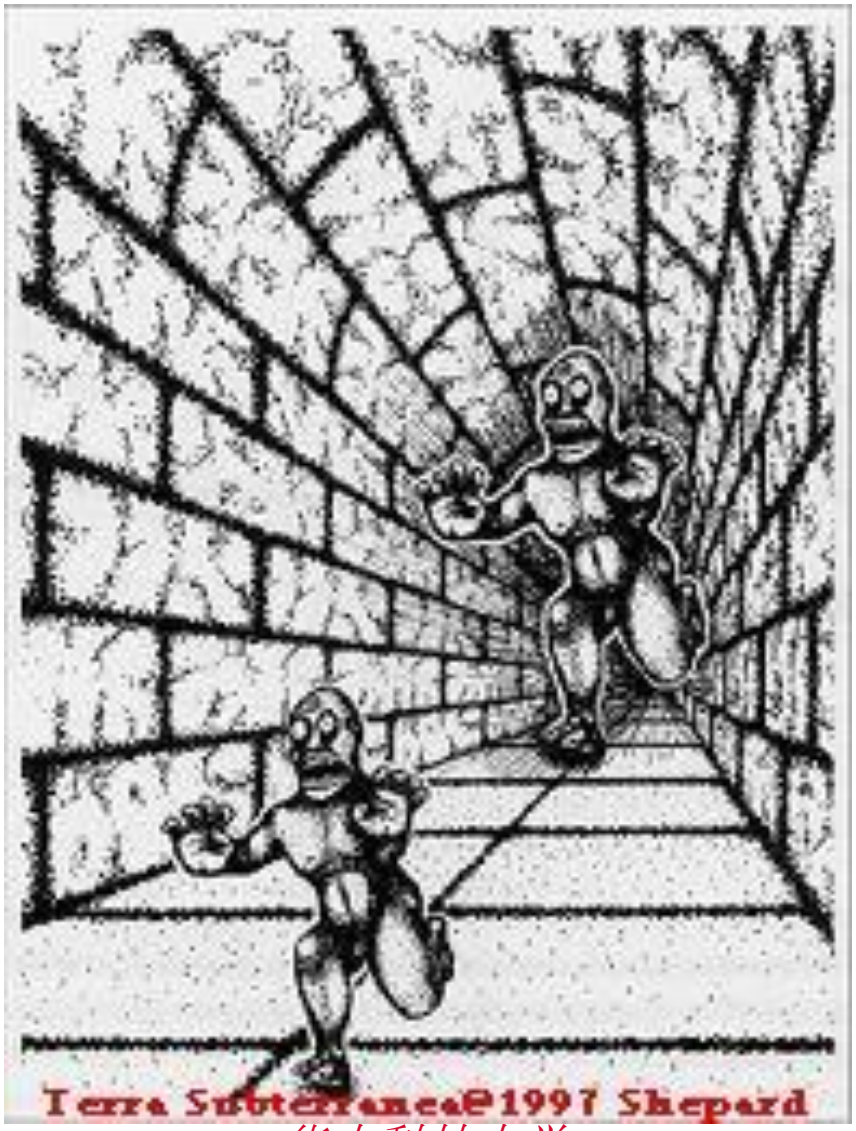
Measuring height



What is the height of the camera?

Slide by Steve Seitz

Perspective cues



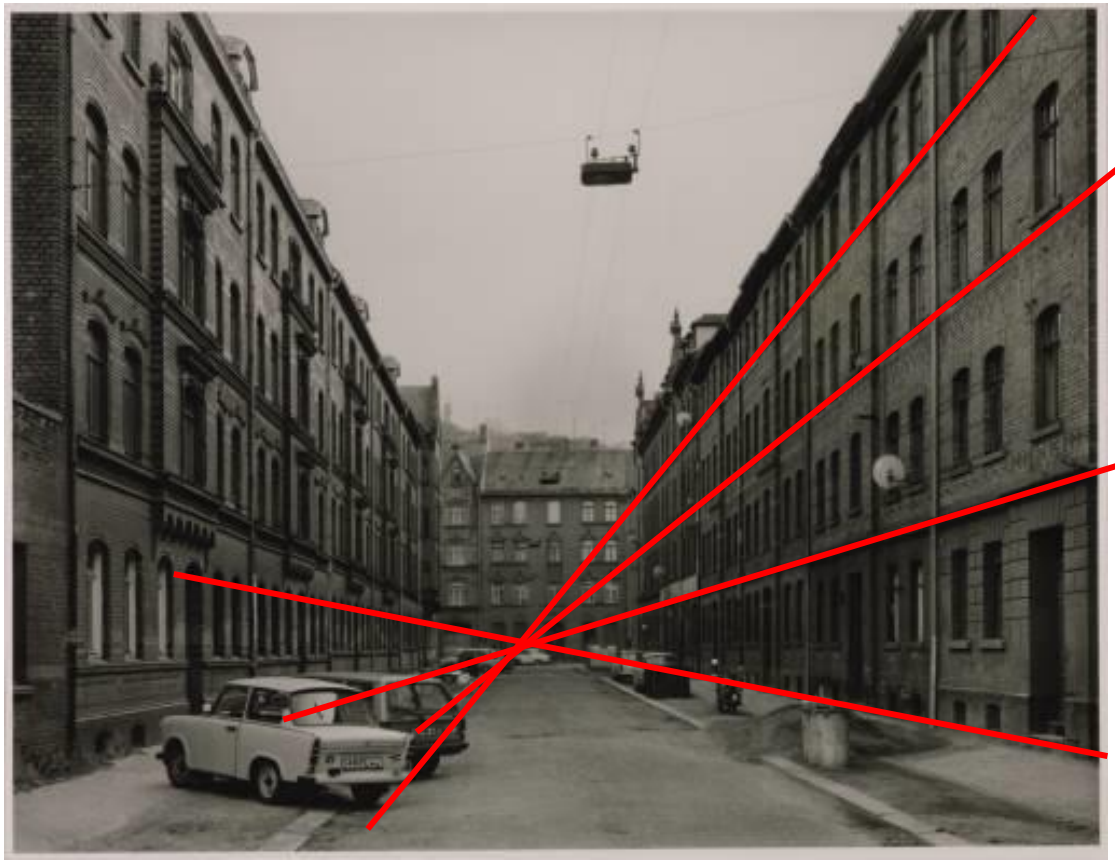
Slide by Steve Seitz

Vanishing points and lines

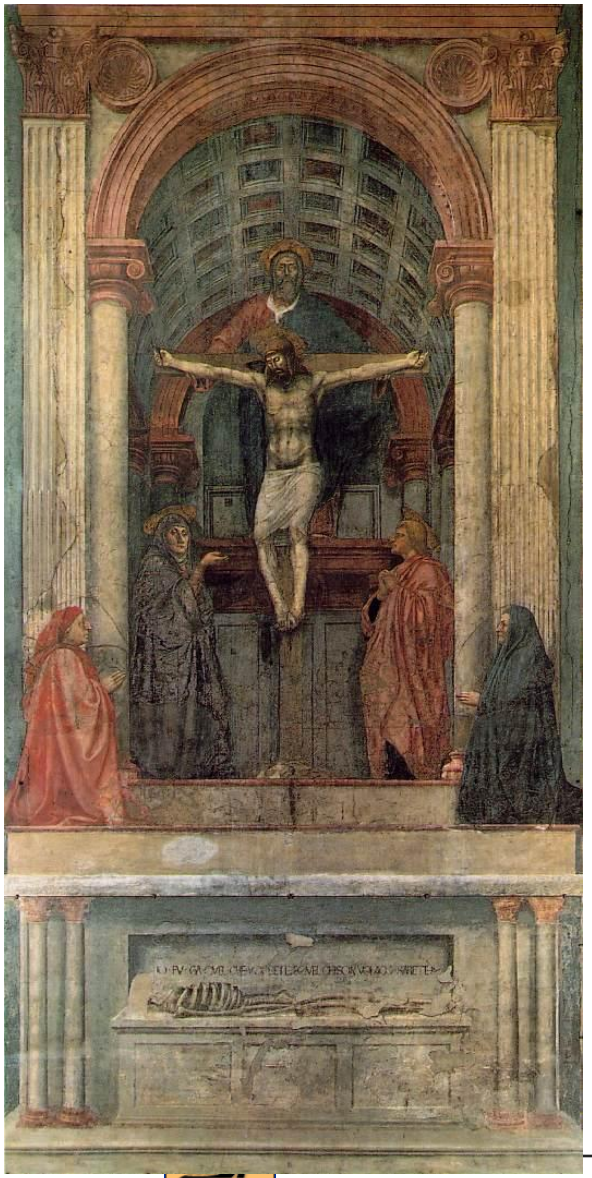


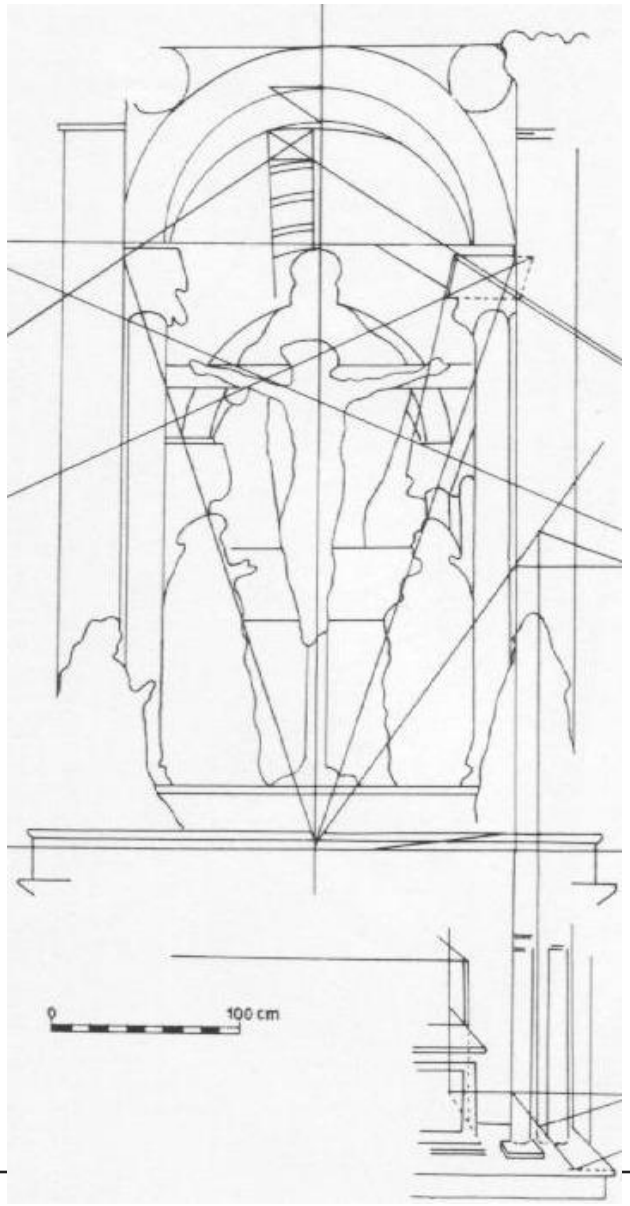
Photo from online Tate collection

Note on estimating vanishing points



One-point perspective





Masaccio, *Trinity*, Santa Maria Novella, Florence, 1425-28

One of the first consistent uses of perspective in Western art

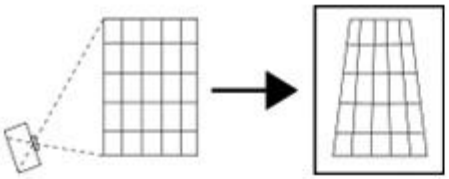
Perspective distortion

- Problem for architectural photography: converging verticals

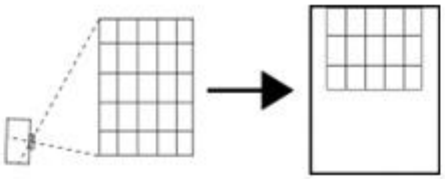


Perspective distortion

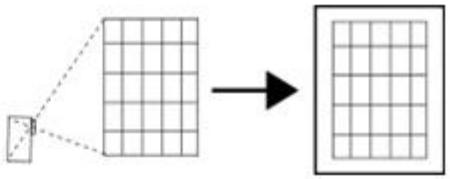
- Problem for architectural photography: converging verticals



Tilting the camera upwards results in converging verticals

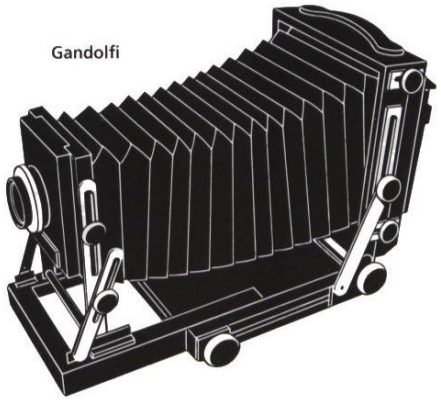


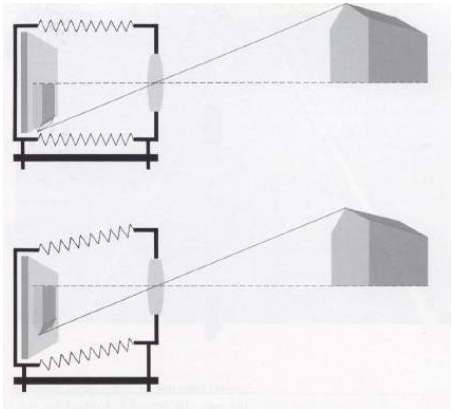
with an ordinary lens, captures only the bottom portion of the building



Shifting the lens upwards results in a picture of the entire subject

- Solution: view camera (lens shifted w.r.t. film)





http://en.wikipedia.org/wiki/Perspective_correction_lens

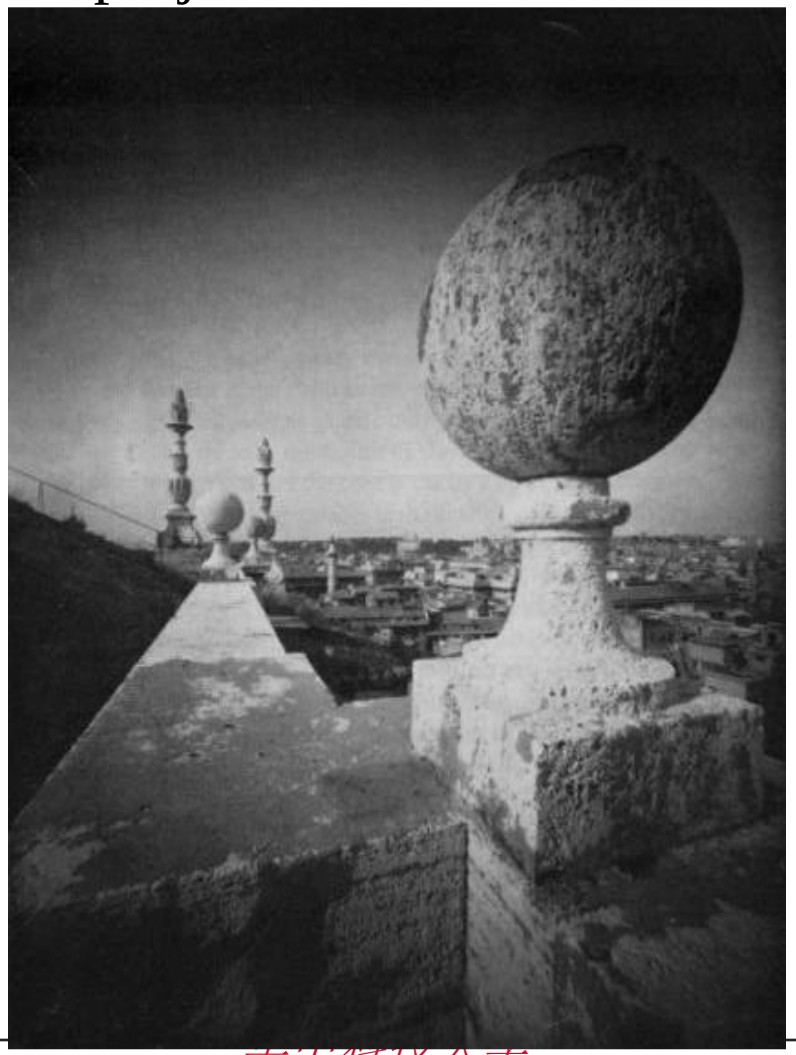
Perspective distortion

- Problem for architectural photography: converging verticals
- Result:



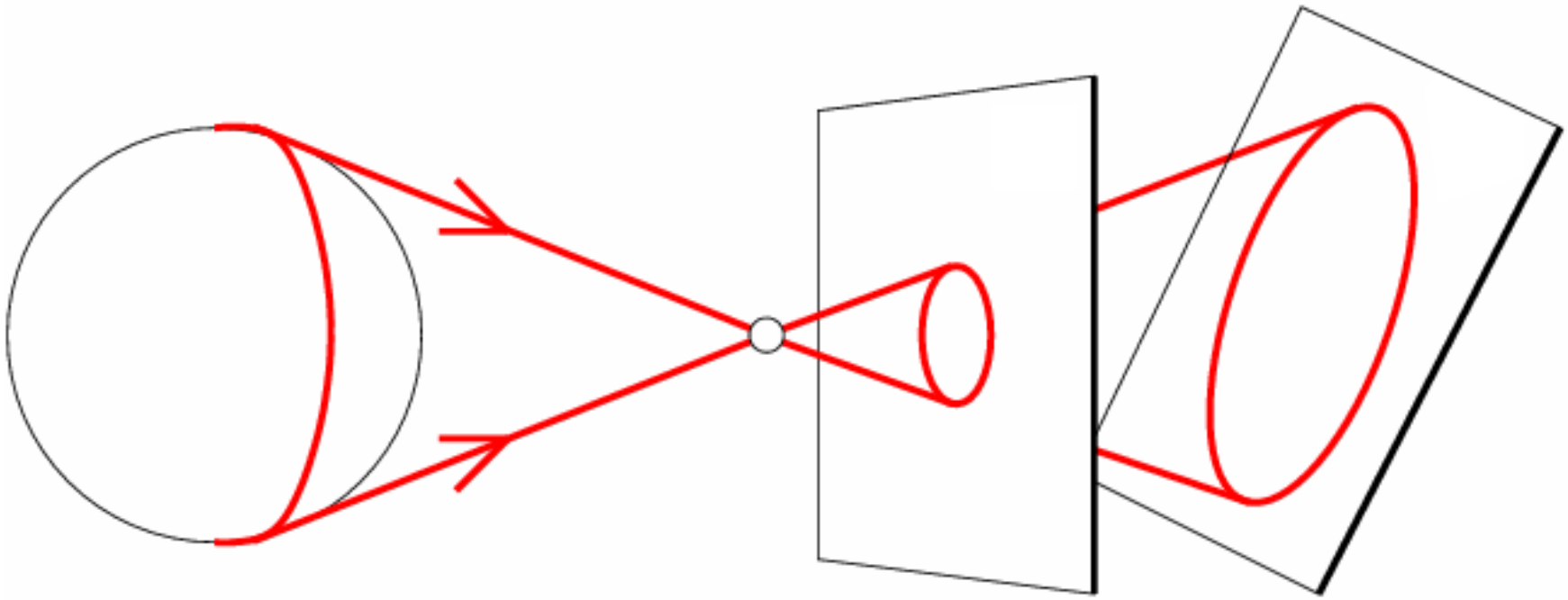
Perspective distortion

- What does a sphere project to?



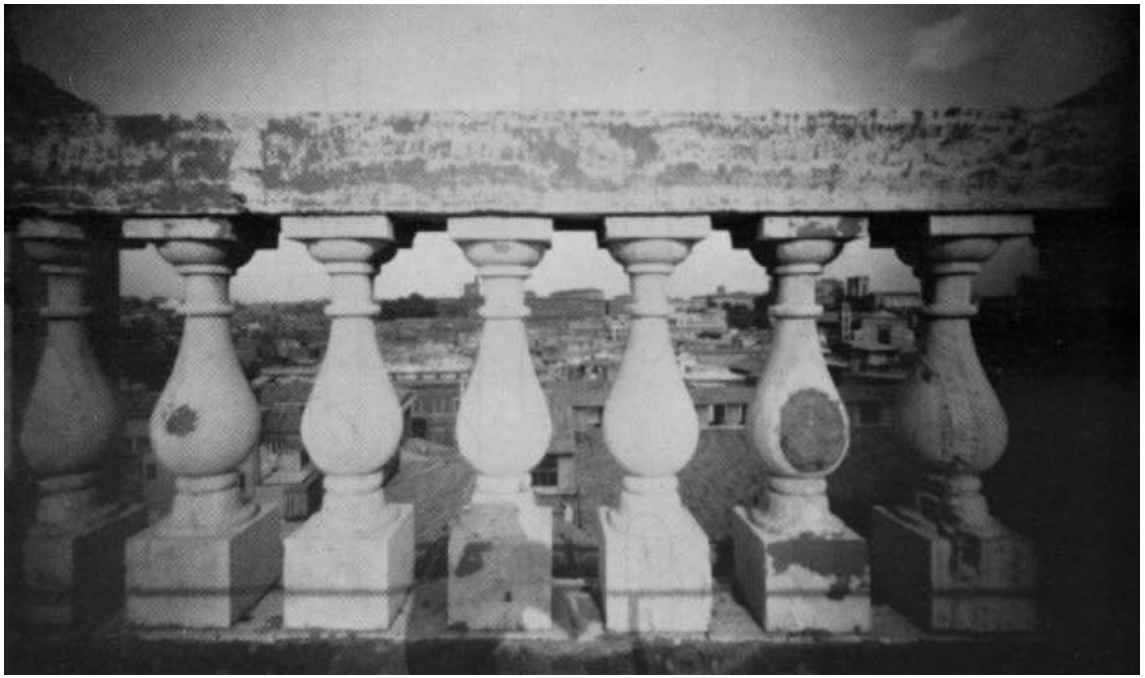
Perspective distortion

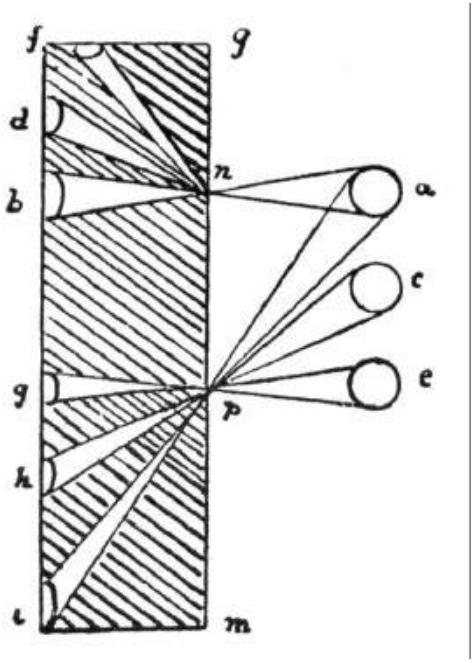
- What does a sphere project to?



Perspective distortion

- The exterior columns appear bigger
- The distortion is not due to lens flaws
- Problem pointed out by Da Vinci

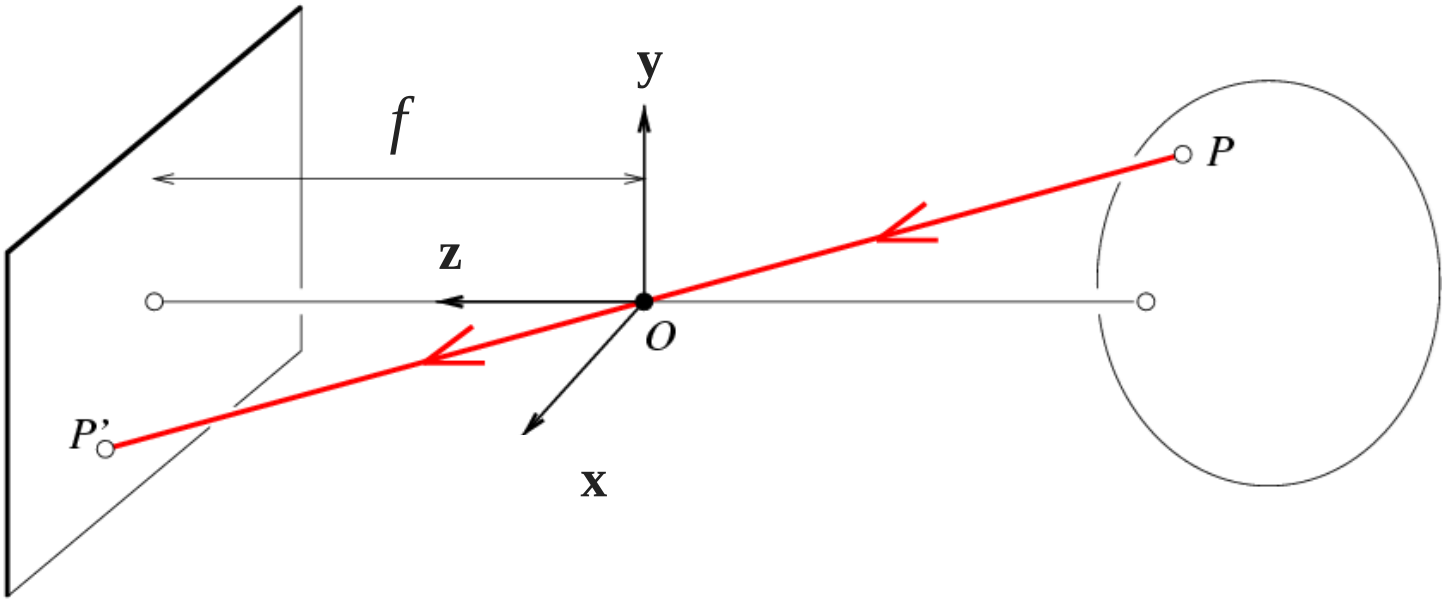




Perspective distortion: People



Modeling projection

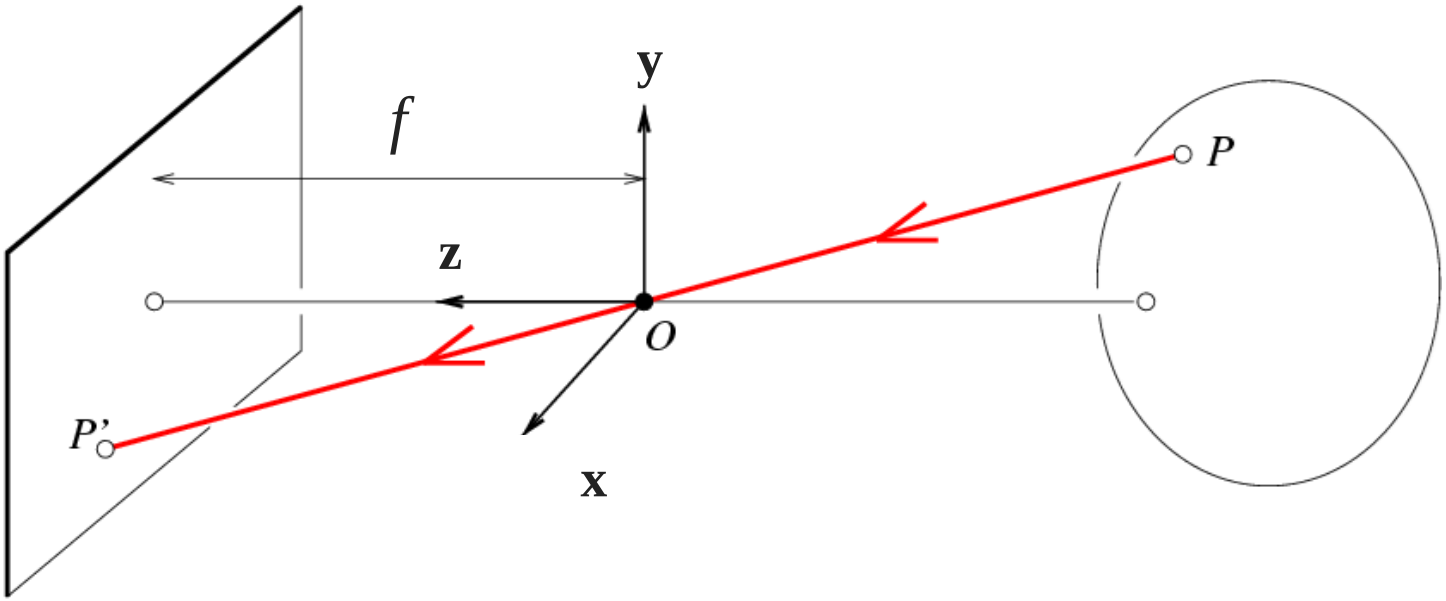


The coordinate system

The optical center (***O***) is at the origin

The image plane is parallel to xy-plane (perpendicular to z axis)

Modeling projection



Projection equations

Compute intersection with image plane of ray from ***P*** = (x,y,z) to ***O***

Derived using similar triangles

$$(x, y, z) \rightarrow (f\frac{x}{z}, f\frac{y}{z}, f)$$

- We get the projection by throwing out the last coordinate:

$$(x, y, z) \rightarrow (f\frac{x}{z}, f\frac{y}{z})$$

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Homogeneous coordinates

$$(x, y, z) \rightarrow (f\frac{x}{z}, f\frac{y}{z})$$

Is this a linear transformation?

- no—division by z is nonlinear

Trick: add one more coordinate:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

homogeneous scene coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Homogeneous coordinates

Invariant to scaling

$$k\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kw \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{kx}{kw} \\ \frac{ky}{kw} \end{bmatrix} = \begin{bmatrix} \frac{x}{w} \\ \frac{y}{w} \end{bmatrix}$$

Homogeneous Coordinates

Cartesian Coordinates

Point in Cartesian is ray in Homogeneous

Basic geometry in homogeneous coordinates

Line equation: ax + by + c = 0

$$line_i = \begin{bmatrix} a_i \\ b_i \\ c_i \end{bmatrix}$$

Append 1 to pixel coordinate to get homogeneous coordinate

Line given by cross product of two points

$$p_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix}$$

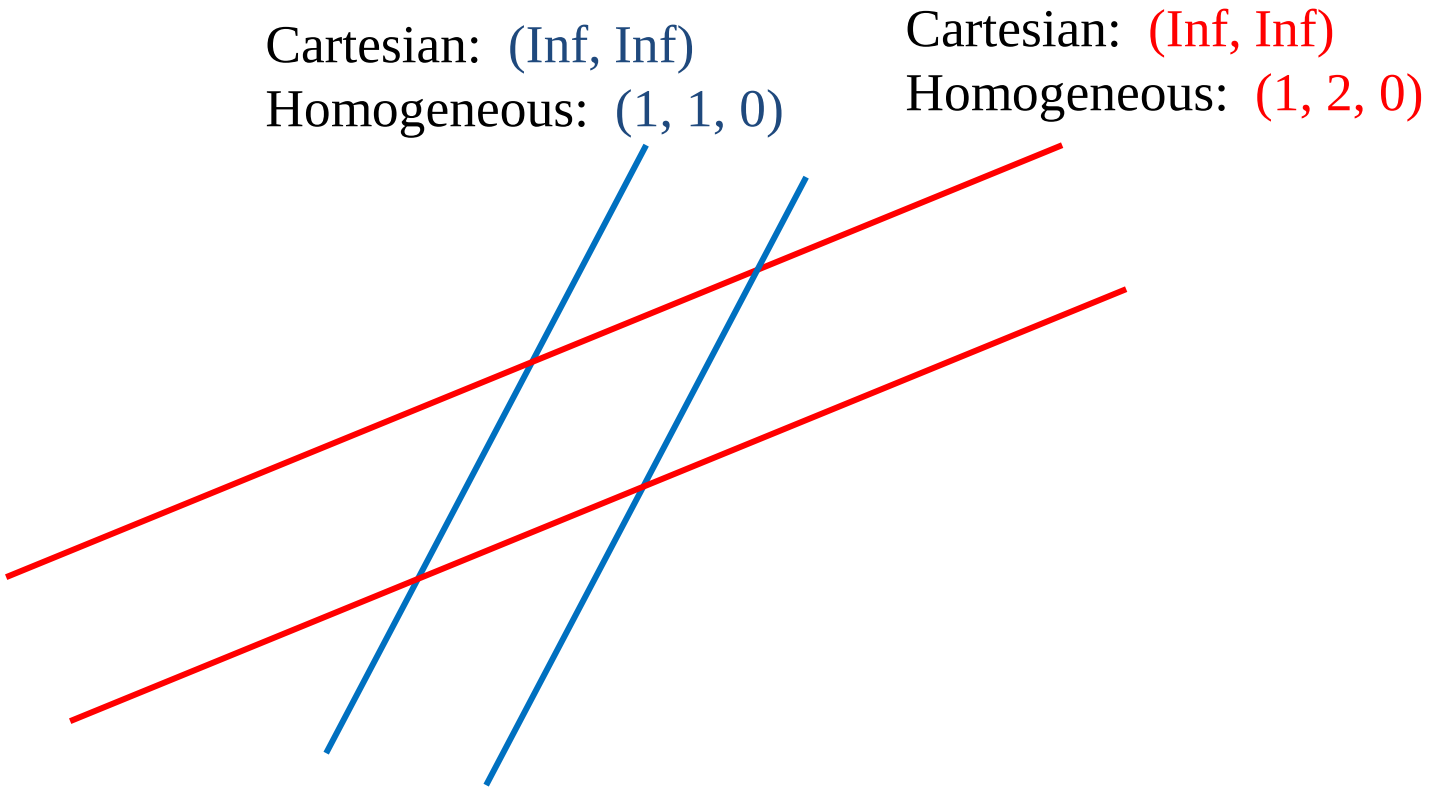
Intersection of two lines given by cross product of the lines

$$line_{ij} = p_i \times p_j$$

$$q_{ij} = line_i \times line_j$$

Another problem solved by homogeneous coordinates

Intersection of parallel lines



Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z/f \end{bmatrix} \Rightarrow (f\frac{x}{z}, f\frac{y}{z})$$

divide by the third coordinate

Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z/f \end{bmatrix} \Rightarrow \left(f\frac{x}{z}, f\frac{y}{z}\right)$$

divide by the third coordinate

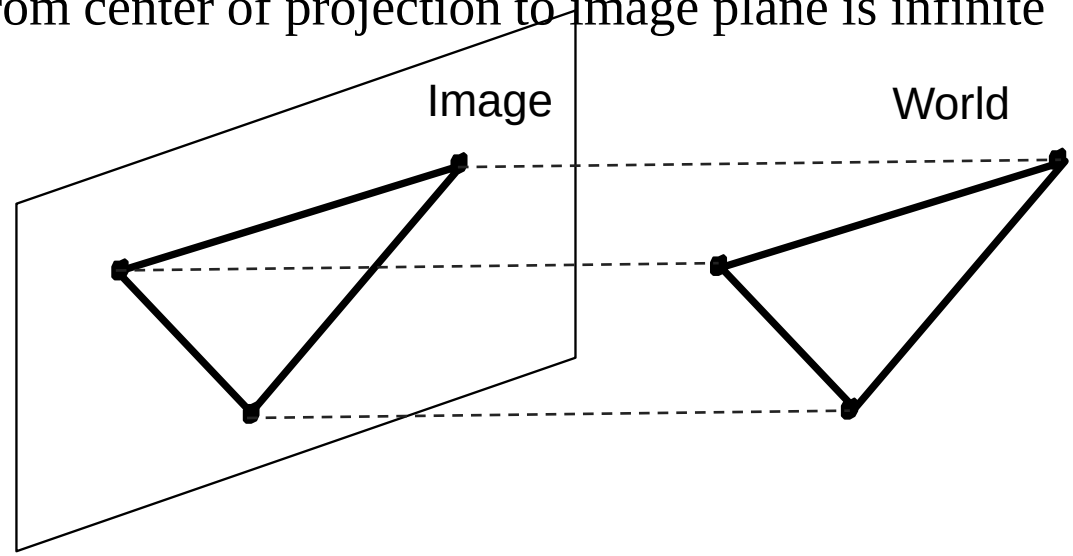
In practice: lots of coordinate transformations…

$$\begin{pmatrix} \text{2D point (3x1)} \end{pmatrix} = \begin{pmatrix} \text{Camera to pixel coord. trans. matrix (3x3)} \end{pmatrix} \begin{pmatrix} \text{Perspective projection matrix (3x4)} \end{pmatrix} \begin{pmatrix} \text{World to camera coord. trans. matrix (4x4)} \end{pmatrix} \begin{pmatrix} \text{3D point (4x1)} \end{pmatrix}$$

Orthographic Projection

Special case of perspective projection

Distance from center of projection to image plane is infinite

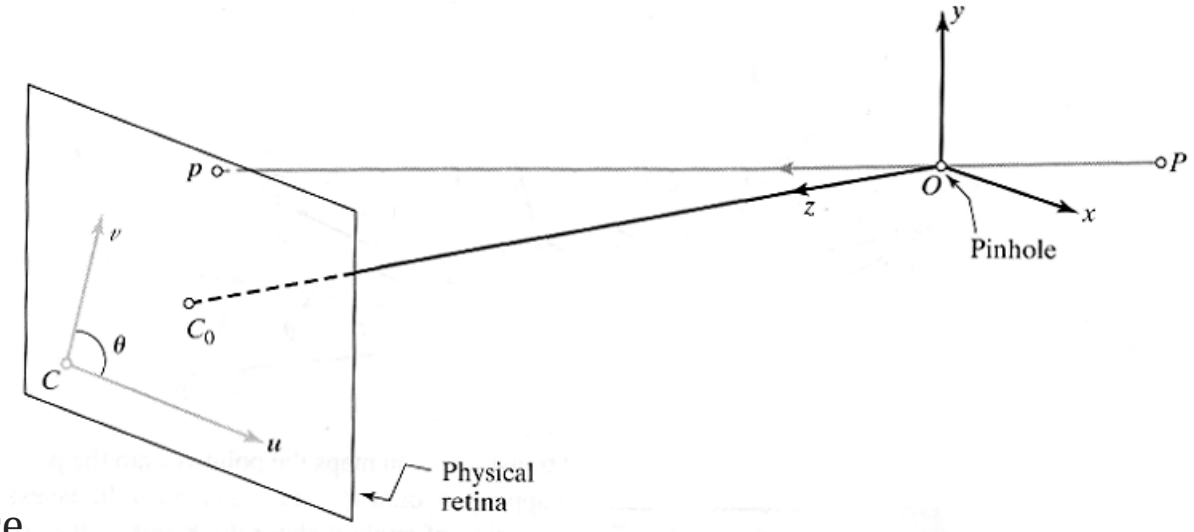


Also called “parallel projection”

What’s the projection matrix?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$

Intrinsic parameters: from idealized world coordinates to pixel values



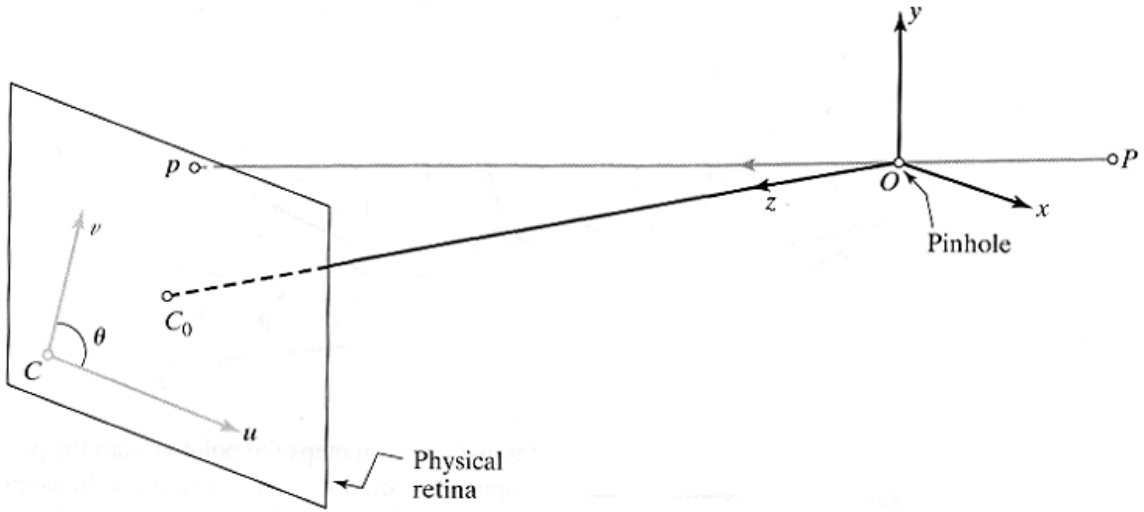
Forsyth&Ponce

Perspective projection

$$u = f\frac{x}{z}$$

$$v = f\frac{y}{z}$$

Intrinsic parameters

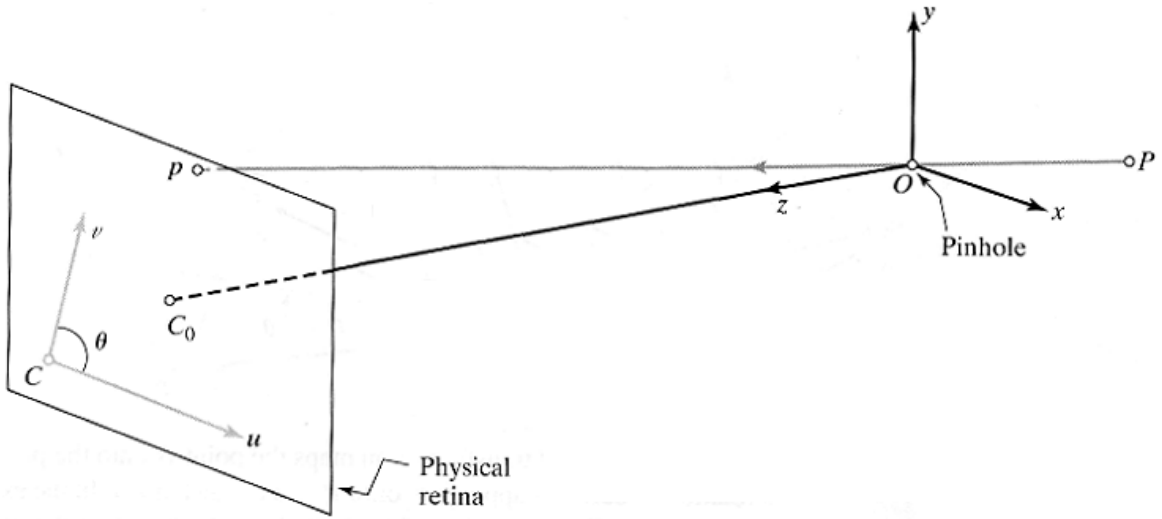


But "pixels" are in some arbitrary spatial units

$$u = \alpha \frac{x}{z}$$

$$v = \alpha \frac{y}{z}$$

Intrinsic parameters

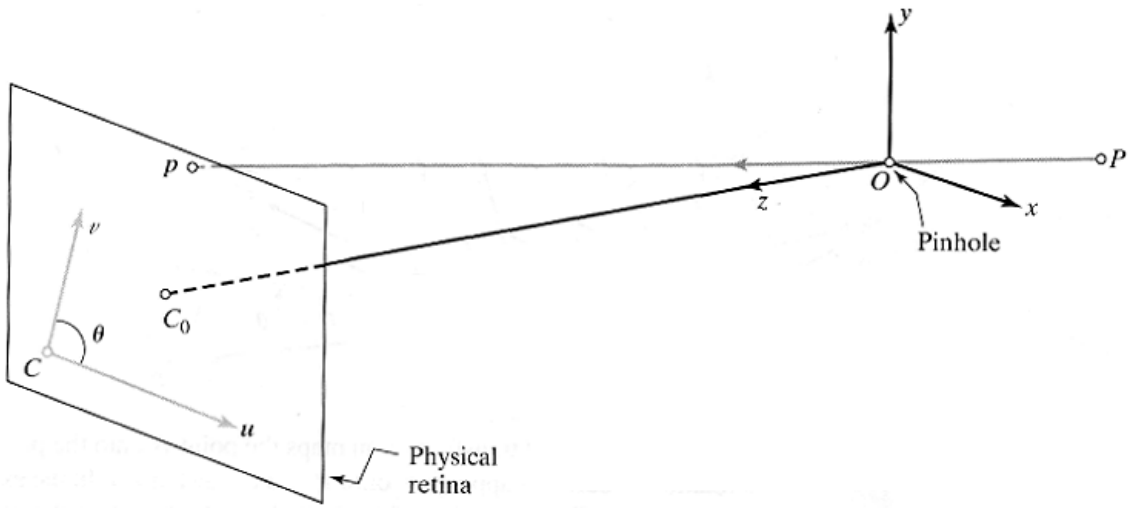


Maybe pixels are not square

$$u = \alpha \frac{x}{z}$$

$$v = \beta \frac{y}{z}$$

Intrinsic parameters

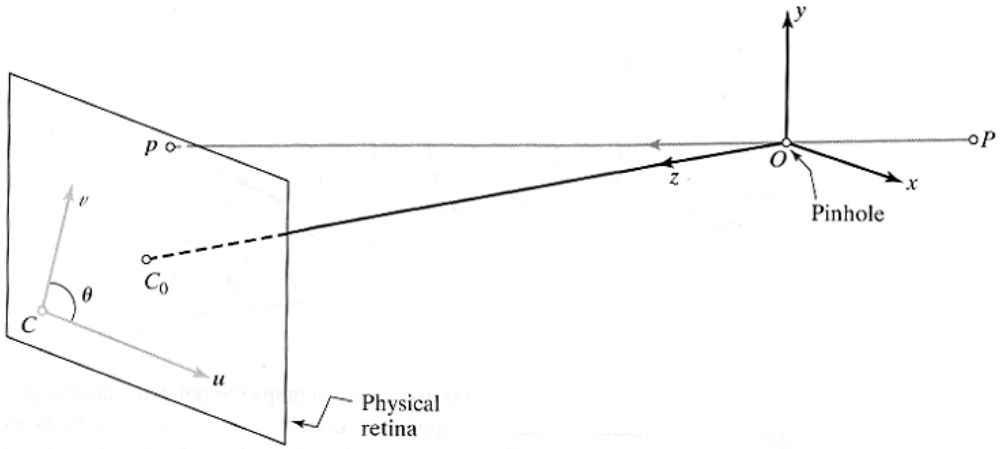


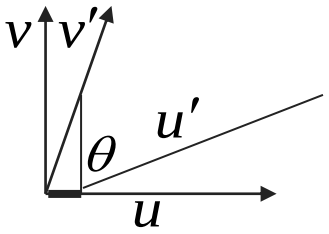
We don't know the origin of our camera pixel coordinates

$$u = \alpha \frac{x}{z} + u_0$$

$$v = \beta \frac{y}{z} + v_0$$

Intrinsic parameters





$$v' \sin(\theta) = v$$

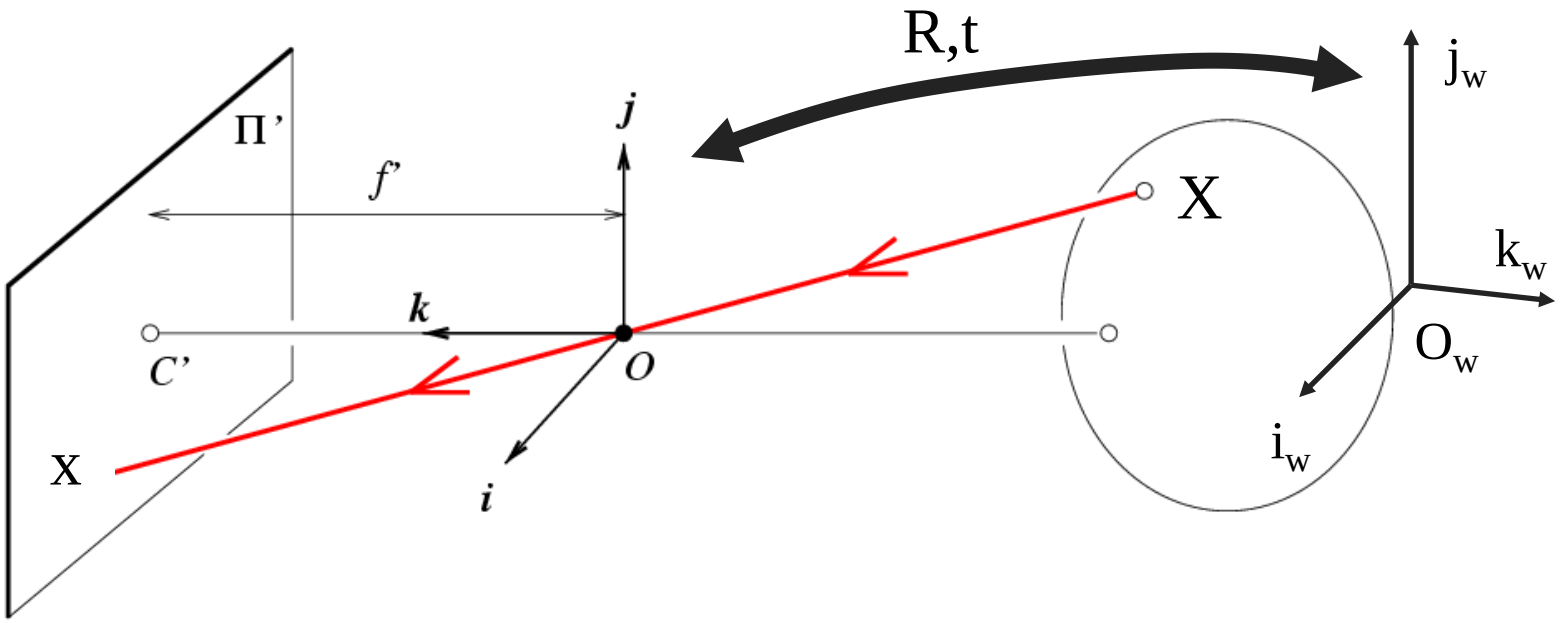
$$u' = u - \cos(\theta) v' = u - \cot(\theta) v$$

May be skew between camera pixel axes

$$u = \alpha \frac{x}{z} - \alpha \cot(\theta) \frac{y}{z} + u_0$$

$$v = \frac{\beta}{\sin(\theta)} \frac{y}{z} + v_0$$

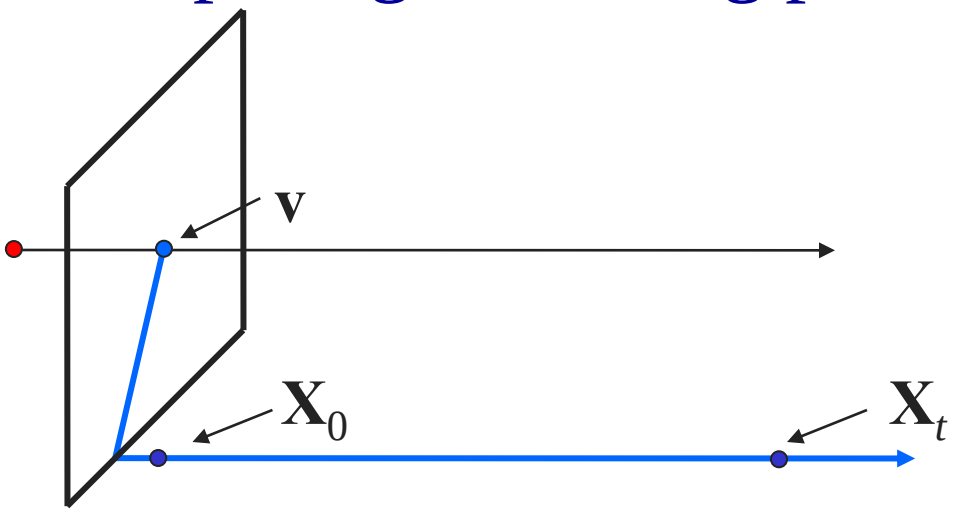
Projection matrix



$$\mathbf{x} = \mathbf{K}\begin{bmatrix}\mathbf{R} & \mathbf{t}\end{bmatrix}\mathbf{X}$$

x: Image Coordinates: (u,v,1)
K: Intrinsic Matrix (3x3)
R: Rotation (3x3)
t: Translation (3x1)
X: World Coordinates: (X,Y,Z,1)

Computing vanishing points



$$\mathbf{X}_t = \begin{bmatrix} x_0 + td_1 \\ y_0 + td_2 \\ z_0 + td_3 \\ 1 \end{bmatrix} = \begin{bmatrix} x_0/t + d_1 \\ y_0/t + d_2 \\ z_0/t + d_3 \\ 1/t \end{bmatrix} \qquad \mathbf{X}_\infty = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ 0 \end{bmatrix}$$

- $\mathbf{X}_\infty$ is a *point at infinity*, $\mathbf{v}$ is its projection: $\mathbf{v} = \mathbf{P}\mathbf{X}_\infty$
- The vanishing point depends only on *line direction*
- All lines having direction $\mathbf{d}$ intersect at $\mathbf{X}_\infty$

Calibration from vanishing points

- Consider a scene with three orthogonal vanishing directions:



- Note: $\mathbf{v}_1$, $\mathbf{v}_2$ are *finite* vanishing points and $\mathbf{v}_3$ is an *infinite* vanishing point

Calibration from vanishing points

- Consider a scene with three orthogonal vanishing directions:



- We can align the world coordinate system with these directions

Calibration from vanishing points

$$\mathbf{P} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 & \mathbf{p}_4 \end{bmatrix}$$

- $\mathbf{p}_1 = \mathbf{P}(1,0,0,0)^T$ – the vanishing point in the x direction
- Similarly, $\mathbf{p}_2$ and $\mathbf{p}_3$ are the vanishing points in the y and z directions
- $\mathbf{p}_4 = \mathbf{P}(0,0,0,1)^T$ – projection of the origin of the world coordinate system
- Problem: we can only know the four columns up to independent scale factors, additional constraints needed to solve for them

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_i \mathbf{v}_i = \mathbf{K}[\mathbf{R} \mid \mathbf{t}] \begin{bmatrix} \mathbf{e}_i \\ 0 \end{bmatrix}$$

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_i \mathbf{v}_i = \mathbf{K}\mathbf{R}\mathbf{e}_i$$

$$\mathbf{e}_i = \lambda_i \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i$$

- Orthogonality constraint:

$$\mathbf{e}_i^T \mathbf{e}_j = 0$$

$$\underbrace{\mathbf{v}_i^T \mathbf{K}^{-T}}_{\mathbf{e}_i^T} \underbrace{\mathbf{R}\mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_j}_{\mathbf{e}_j} = 0$$

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_i \mathbf{v}_i = \mathbf{K}\mathbf{R}\mathbf{e}_i$$

$$\mathbf{e}_i = \lambda_i \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i$$

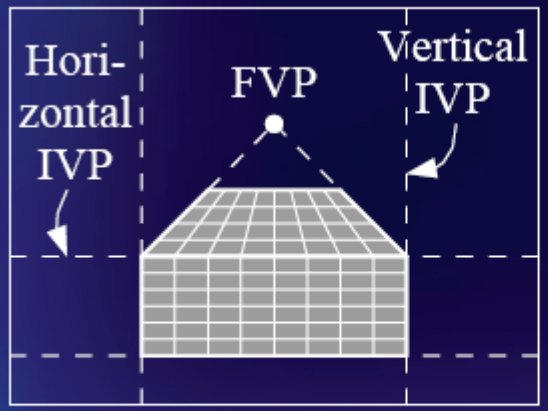
- Orthogonality constraint:

$$\mathbf{e}_i^T \mathbf{e}_j = 0$$

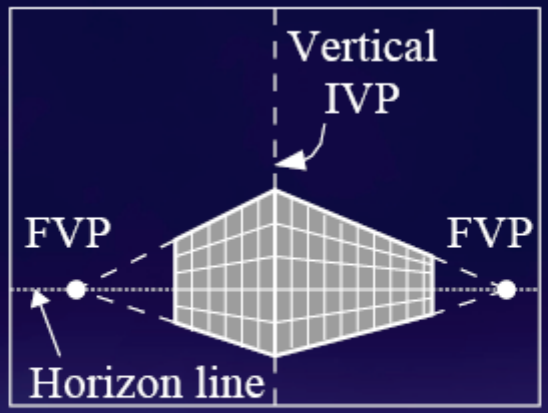
$$\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{v}_j = 0$$

- Rotation disappears, each pair of vanishing points gives constraint on focal length and principal point

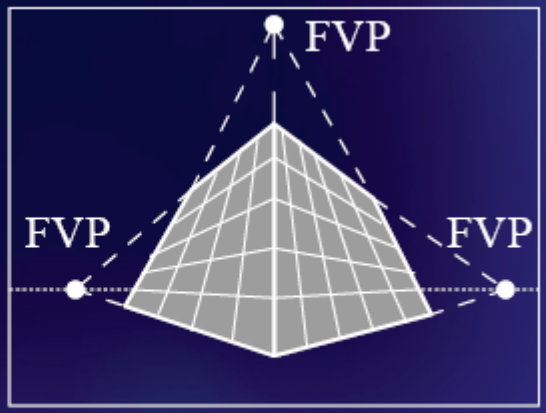
Calibration from vanishing points



1 finite vanishing point,
2 infinite vanishing points



2 finite vanishing points,
1 infinite vanishing point

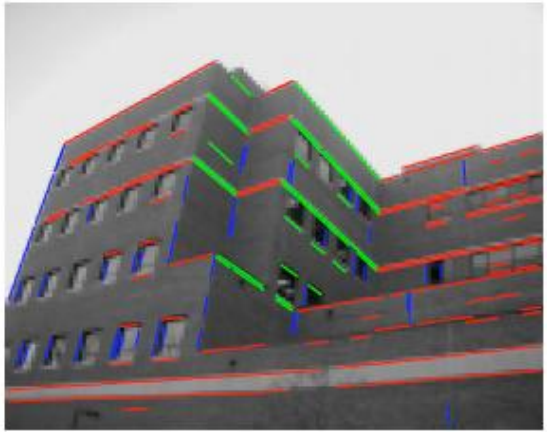


3 finite vanishing points



Cannot recover focal length, principal point is the third vanishing point





Can solve for focal length, principal point

Rotation from vanishing points

- Constraints on vanishing points: $\lambda_i \mathbf{v}_i = \mathbf{K}\mathbf{R}\mathbf{e}_i$
- After solving for the calibration matrix:

$$\lambda_i \mathbf{K}^{-1}\mathbf{v}_i = \mathbf{R}\mathbf{e}_i$$

- Notice:

$$\mathbf{R}\mathbf{e}_1 = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{r}_1$$

- Thus,

$$\mathbf{r}_i = \lambda_i \mathbf{K}^{-1}\mathbf{v}_i$$

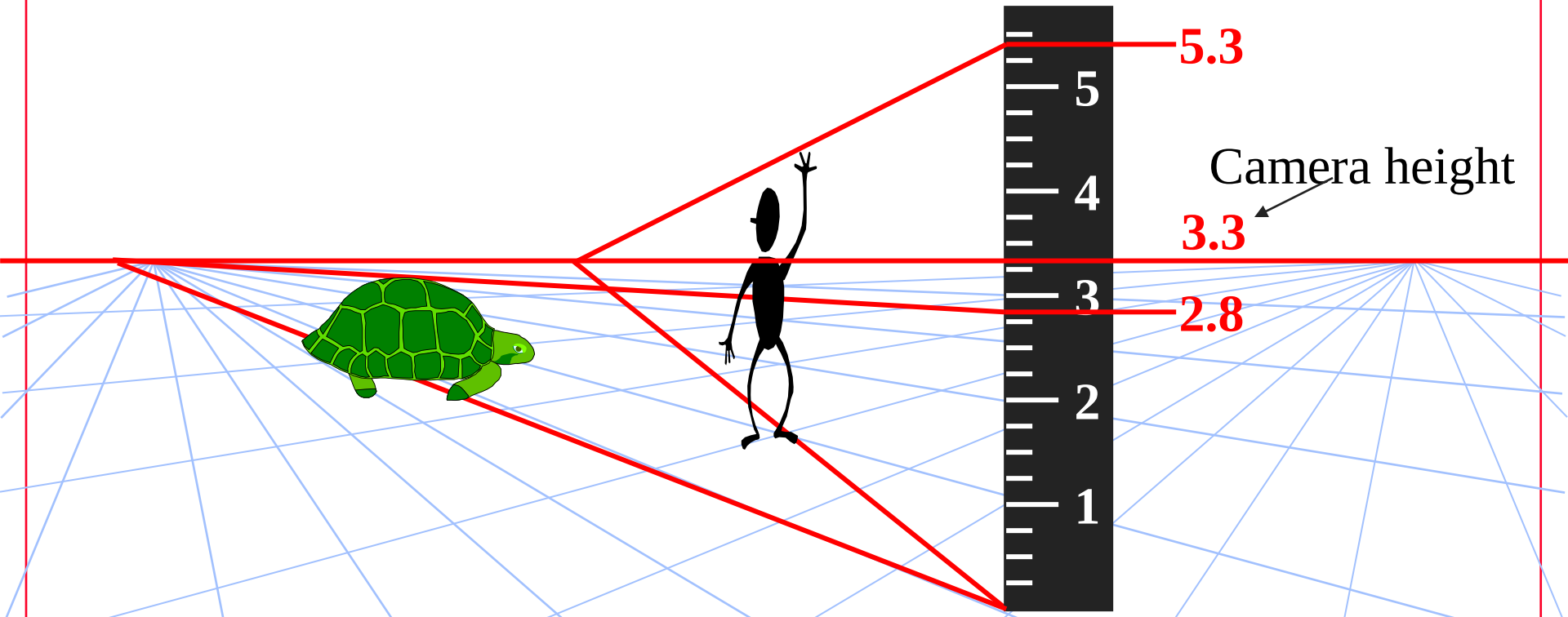
- Get λ_i by using the constraint $\|\mathbf{r}_i\|^2 = 1$.

Calibration from vanishing points: Summary

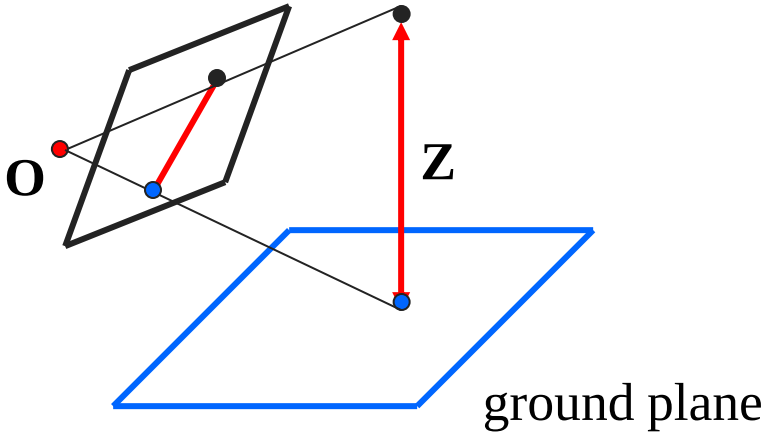
- Solve for K (focal length, principal point) using three orthogonal vanishing points
- Get rotation directly from vanishing points once calibration matrix is known

- Advantages
  - No need for calibration chart, 2D-3D correspondences
  - Could be completely automatic
- Disadvantages
  - Only applies to certain kinds of scenes
  - Inaccuracies in computation of vanishing points
  - Problems due to infinite vanishing points

Recall: Measuring height



Measuring height without a ruler



Compute Z from image measurements

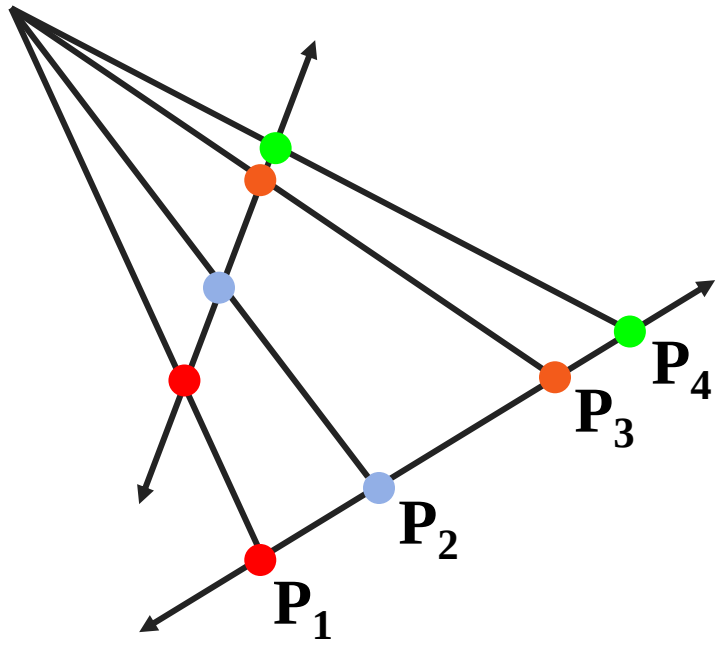
- Need more than vanishing points to do this

Projective invariant

- We need to use a *projective invariant*: a quantity that does not change under projective transformations (including perspective projection)
  - What are some invariants for similarity, affine transformations?

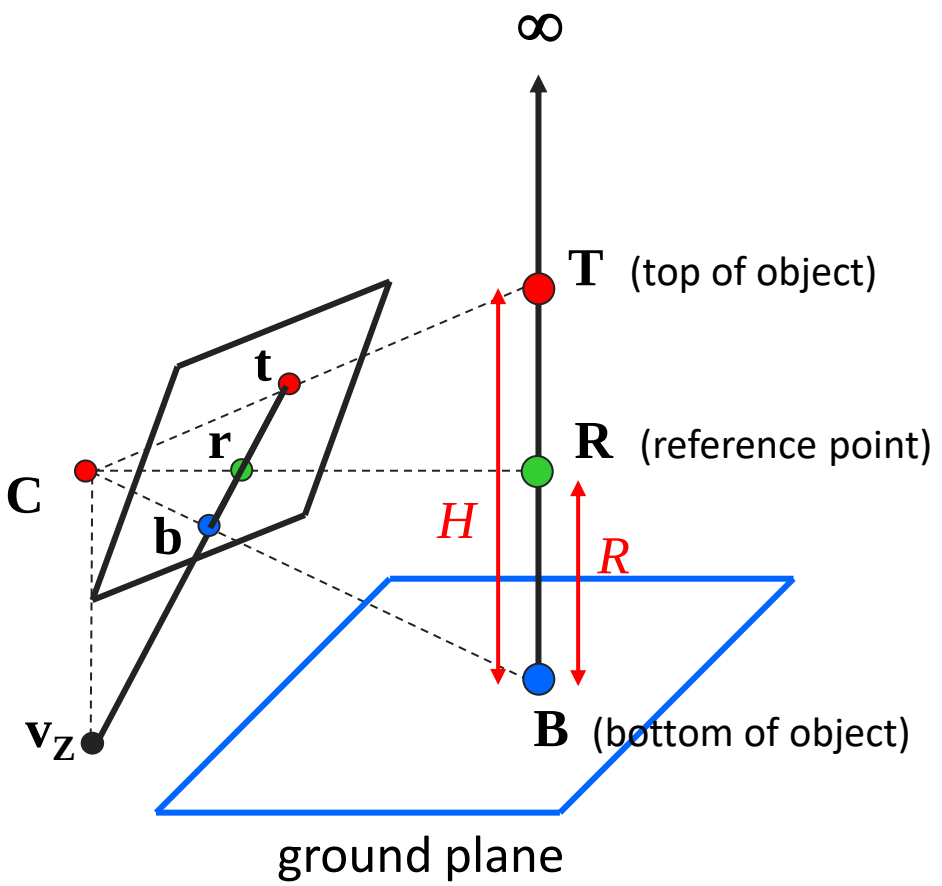
Projective invariant

- We need to use a *projective invariant*: a quantity that does not change under projective transformations (including perspective projection)
- The *cross-ratio* of four points:



$$\frac{\|\mathbf{P}_3 - \mathbf{P}_1\| \, \|\mathbf{P}_4 - \mathbf{P}_2\|}{\|\mathbf{P}_3 - \mathbf{P}_2\| \, \|\mathbf{P}_4 - \mathbf{P}_1\|}$$

Measuring height



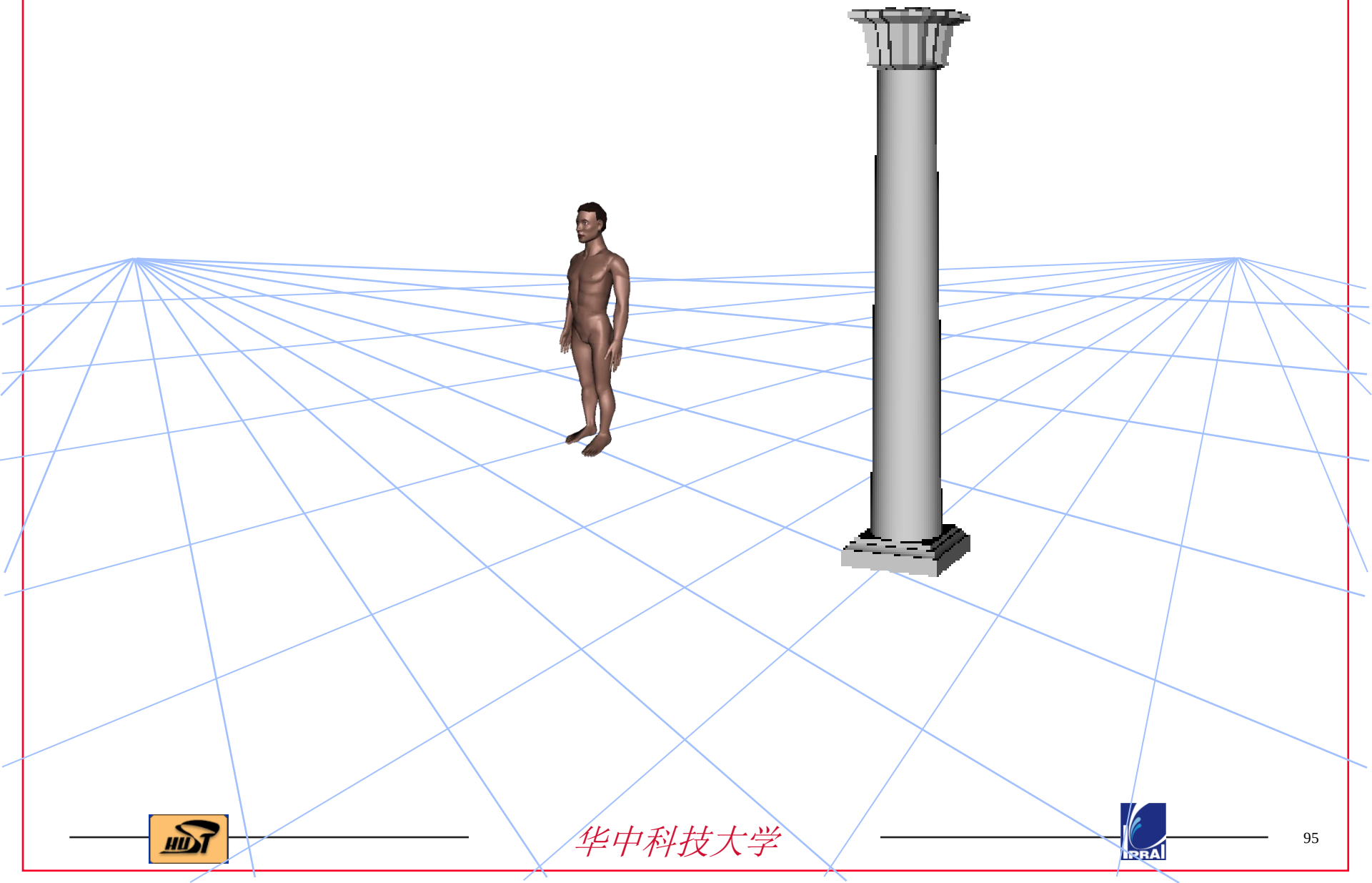
$$\frac{\|\mathbf{T}-\mathbf{B}\|\,\|\infty-\mathbf{R}\|}{\|\mathbf{R}-\mathbf{B}\|\,\|\infty-\mathbf{T}\|}=\frac{H}{R}$$

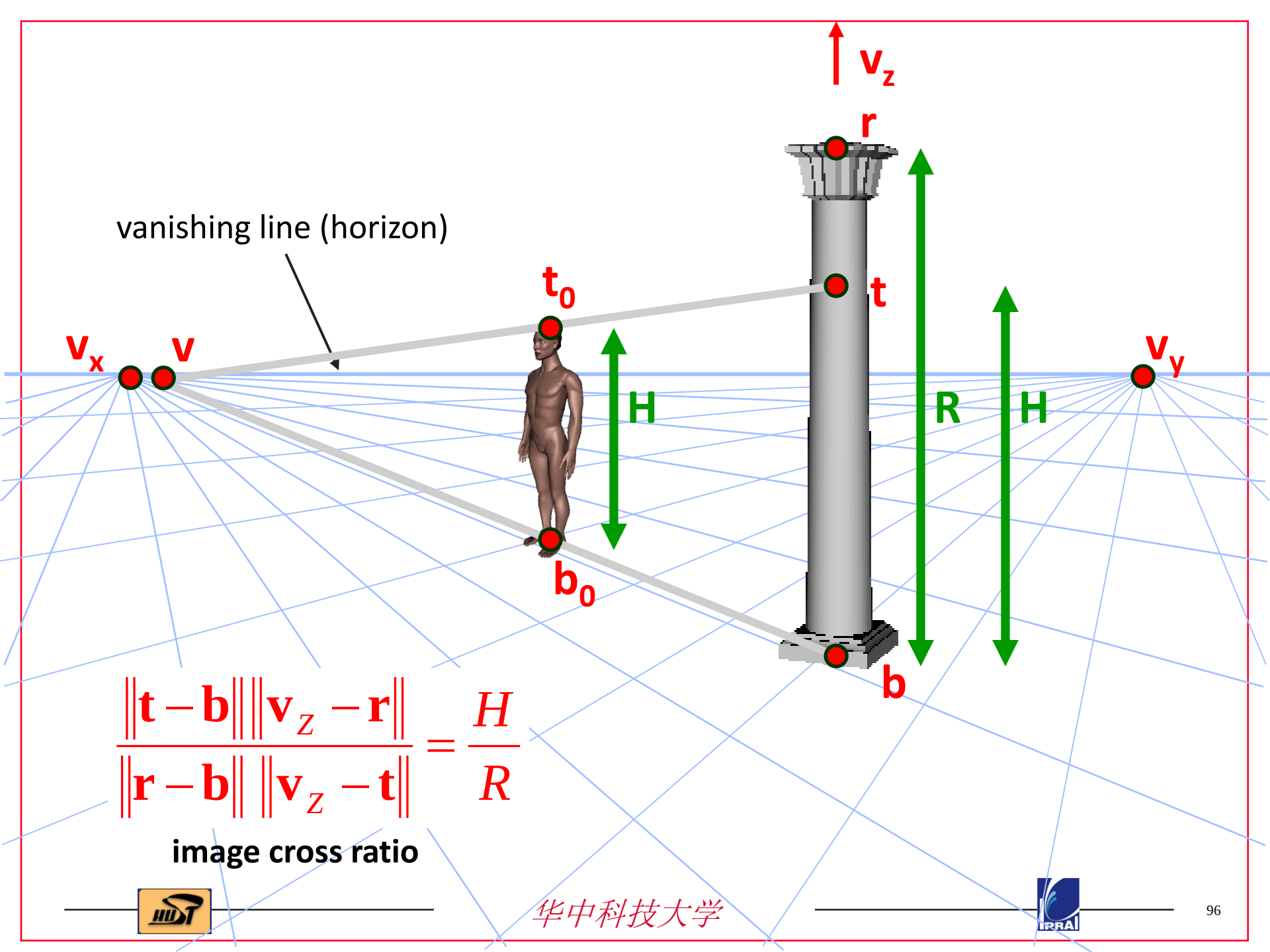
scene cross ratio

$$\frac{\|\mathbf{t}-\mathbf{b}\|\,\|\mathbf{v}_Z-\mathbf{r}\|}{\|\mathbf{r}-\mathbf{b}\|\,\|\mathbf{v}_Z-\mathbf{t}\|}=\frac{H}{R}$$

image cross ratio

Measuring height without a ruler





$$\frac{\|\mathbf{t}-\mathbf{b}\|\|\mathbf{v}_Z-\mathbf{r}\|}{\|\mathbf{r}-\mathbf{b}\|\|\mathbf{v}_Z-\mathbf{t}\|}=\frac{H}{R}$$

image cross ratio

2D lines in homogeneous coordinates

- Line equation: $ax + by + c = 0$

$$\mathbf{l}^T\mathbf{x} = 0 \quad \text{where} \quad \mathbf{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

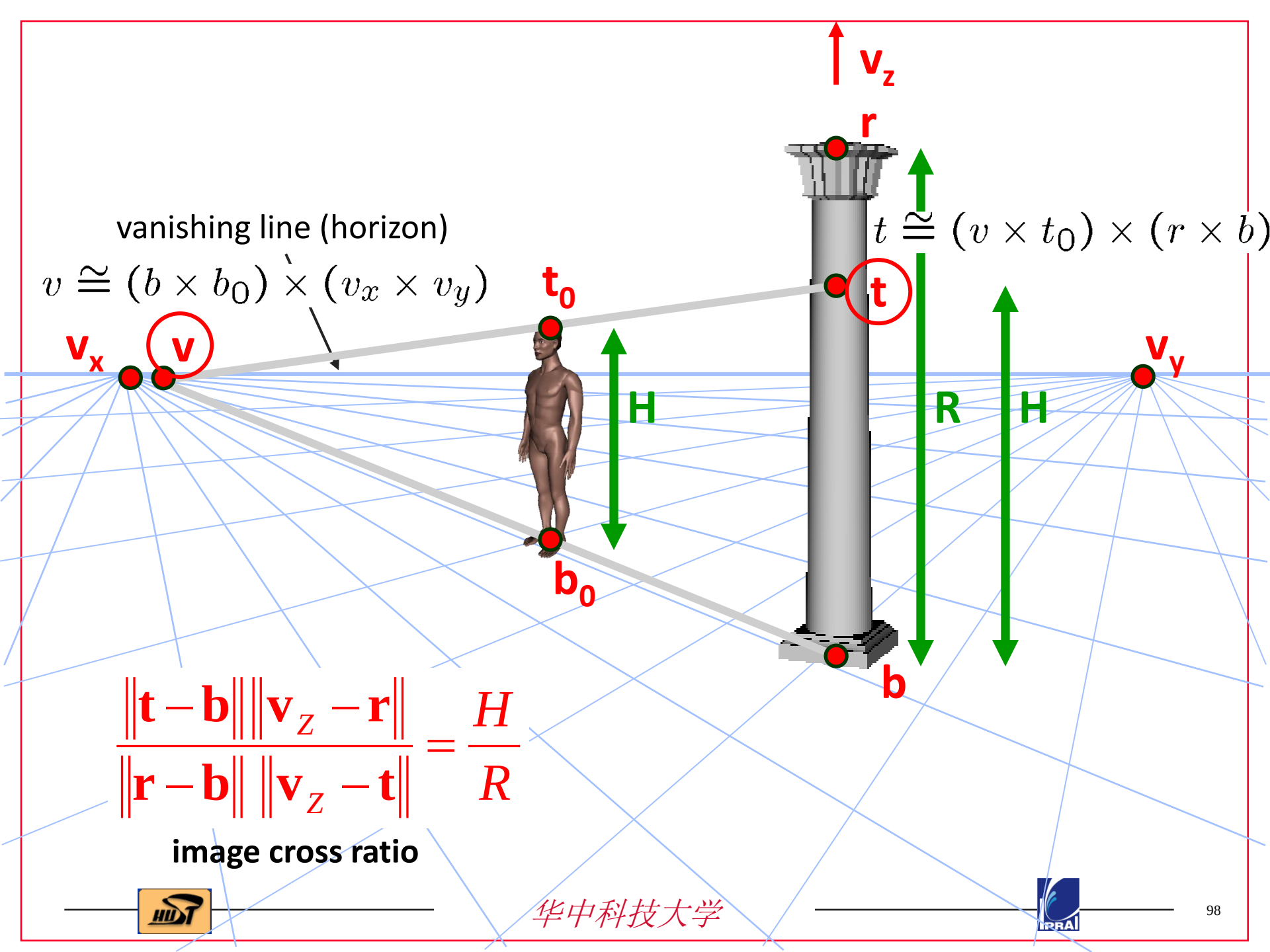
- Line passing through two points:

$$\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$$

- Intersection of two lines:

$$\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$$

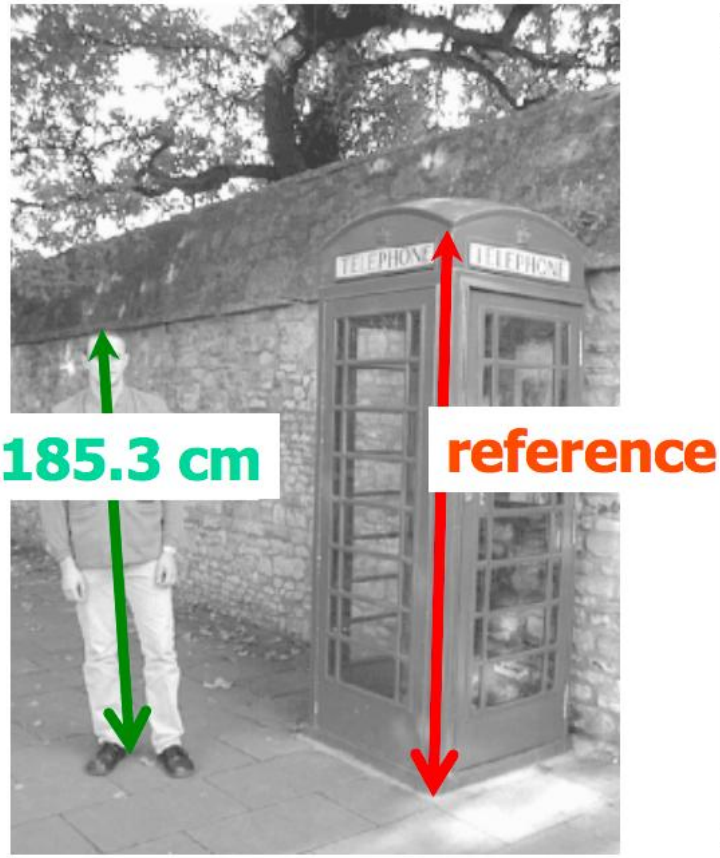
  - What is the intersection of two parallel lines?

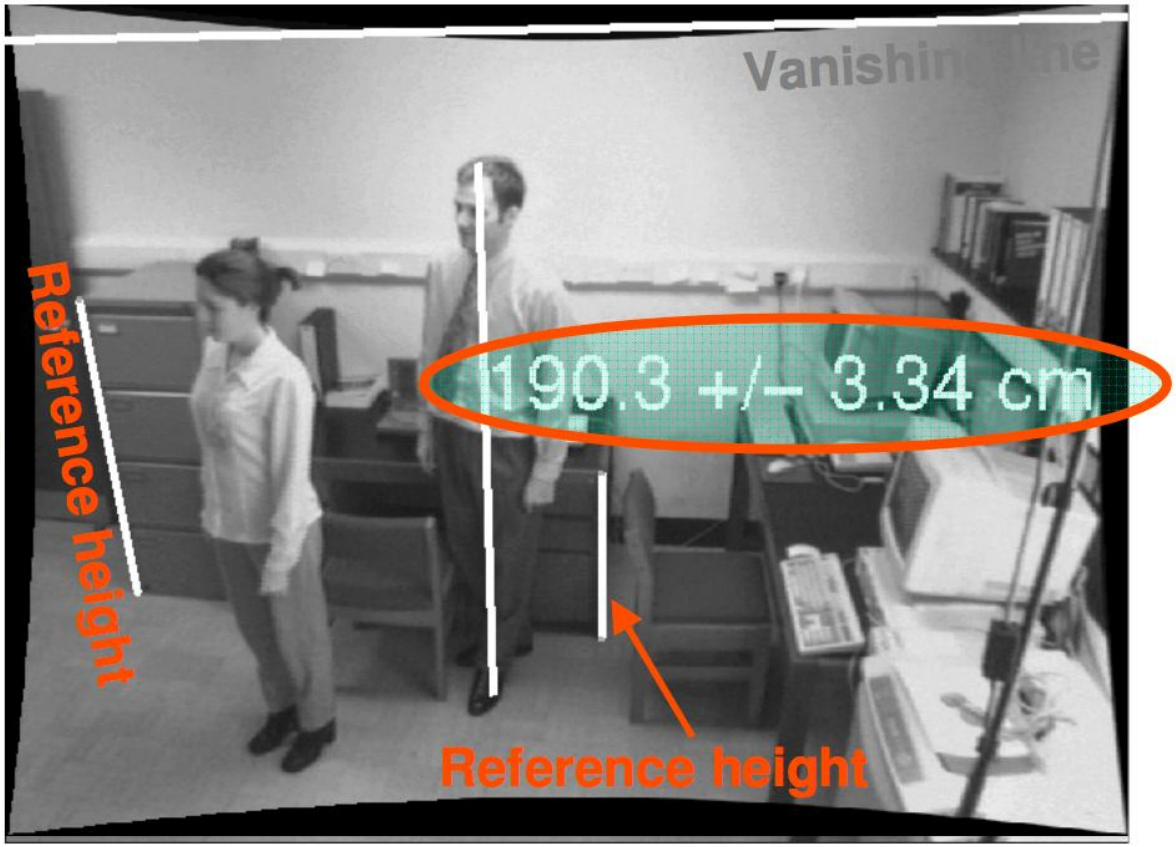


$$\frac{\|\mathbf{t}-\mathbf{b}\|\|\mathbf{v}_Z-\mathbf{r}\|}{\|\mathbf{r}-\mathbf{b}\|\|\mathbf{v}_Z-\mathbf{t}\|}=\frac{H}{R}$$

image cross ratio

Examples





A. Criminisi, I. Reid, and A. Zisserman, Single View Metrology, IJCV 2000

Figure from UPenn CIS580 slides

华中科技大学

Inserting photographed objects into images (Photo clip art，SIGGRAPH 2007)



Original



Created